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Discovery of Molecular Vulnerabilities in Multiple Myeloma for Drug Discovery

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Discovery of Molecular Vulnerabilities in Multiple Myeloma for Drug Discovery

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LIST OF ABBRIVIATIONS

Abbreviation	Explanation
BIC	Bayesian Information Criterion
CCL	Cancer Cell Line Encyclopedia
CRISPR	Clustered Regularly Interspaced Short Palindromic Repeats
DAVID	Database for Annotation, Visualization, and Integrated Discovery
DepMap	Cancer Dependency Map
FDR	False Discovery Rate
GDR	Gene Dependency Relationship
GMM	Gaussian Mixture Modelling
gRNA	Guide RNA
KEGG	Kyoto Encyclopedia of Genes and Genomes
LoF	Loss-of-Function
MM	Multiple Myeloma
MMRF	Multiple Myeloma Research Foundation
mRNA	Messenger RNA
NRM	Network Regulatory Motif
PI	Proteasome Inhibitors
PARP	poly-ADP-ribose polymerase inhibitors
RISC	RNA-Induced Silencing Complex
RNAi	RNA interference
sgRNA	Single Guide RNA
shRNA	Short Hairpin RNA
siRNA	Small Interfering RNA
SL	Synthetic Lethality
UPS	Ubiquitin Proteasome System

ABSTRACT

Introduction

Multiple myeloma (MM) is a blood cancer that develops in plasma cells in the bone marrow which produce monoclonal immunoglobulins (M-protein) that can accumulate in vital organs and cause their dysfunction. Despite advancements from novel therapeutic strategies, multiple myeloma remains incurable, with some disease subgroups unresponsive for the most recent therapy approaches. One of the promising ways to overcome this issue is a search for molecular vulnerabilities which are conferred to cancer cells by mutations. Mutations in cancer cells contributing to their development also confer them specific vulnerabilities that do not present in normal cells. Thus, cancer genetic alterations can be leveraged to find these molecular vulnerabilities called Achilles' Heels, which then aids identifying potential drug targets. Gene dependencies resulting from the tumour biology pathological functioning follow certain regulations which can be indicated by transcriptional network regulatory motifs.

The aim of the present study was to find genes and gene pairs involved in the prevalent gene dependency relationships (GDR) type and identify the category of this GRD in present collection of MM samples through transcription network regulatory motifs (NRM) determination. Initially, the assumed GDR was synthetic lethality. The suspected biological system involving analysed genes was ubiquitin-proteasome pathway.

Methods

A dataset taken from Multiple myeloma Research Foundation (MMRF) site and previously treated with MultiSEp approach was analyzed using R language for dataset transformation and sifting. The most important cluster combinations in gene pair clustering and depletion patterns of gene expression tables were found to reveal transcriptional regulatory motifs. Gene pairs and genes were extracted for further functional biological analysis.

Results

The trend of the pattern configuration showed mutual inhibition closest to the negative feedback loop regulation network. The prevalence of gene pairs generating three or four clusters was detected. The distribution of single gene counts appeared to indirectly indicate the strength of the relationship between gene pairs of a pattern.

Conclusion

Though the tendency within the patterns and corresponding NRM was found, further study is necessary to validate the present analysis findings and to finalize the gene ontology enrichment and functional annotation. This allows constructing a network of biological processes provided by the studied genes and precise the GDRs they are implicated in MM.

Keywords

Multiple myeloma, gene dependency relationships, network regulatory motif, synthetic lethality, pattern, molecular vulnerabilities, drug discovery.

I. INTRODUCTION

Cancer development follows common underlying principles across different types due to the disruption of fundamental biological processes like cell proliferation, differentiation, and death. These disruptions grant tumour cells molecular, biochemical, and cellular traits that define their malignant nature, traits shared by most human cancers. Such changes alter the biology of the entire organism, affecting metabolism and immune responses to sustain tumour growth. (Hanahan, 2000; Hanahan, 2011)

The capabilities that cell acquires to become a cancer cell are described in one of the most important cancer articles, “Hallmark of Cancer”, 2000. They include self-sufficiency in growth signals, insensitivity to anti-growth signals, evading apoptosis, limitless replicative potential, sustainable angiogenesis, tissue invasion and metastasis. (Hanahan, 2000) In the second version of the article issued in 2011 two emerging capabilities were added – metabolic reprogramming and avoiding immune destruction, and two enabling characteristics – genome instability and mutability and inflammation (Hanahan, 2011). The latest issue of the article describes four new hallmarks: unlocking phenotypic plasticity, nonmutational epigenetic reprogramming, polymorphic microbiomes and senescent cells. (Hanahan, 2022)

Cancer cell development and tumour progression rely on two key factors. First, multiple cellular defence mechanisms must be disrupted for a cancer cell to emerge. Second, the normal cell must undergo a series of physiological changes to acquire the characteristics necessary for tumour progression. These processes ultimately result in the cancer cell's autonomy. (Hanahan, 2000)

The initiation of cancer is primarily driven by mutations in two key gene categories: oncogenes, which gain function, and tumour suppressor genes, which lose function. These genetic alterations disrupt normal cellular regulation, leading to uncontrolled growth and tumour formation. (Hanahan, 2000) However, there is also evidence that non-mutational epigenetic reprogramming and polymorphic microbiomes play a role in the development of cancer cell traits (Hanahan, 2022). The tumour microenvironment, along with senescent cells, is crucial for sustaining the vital activity of cancer cells (Hanahan, 2011; Hanahan, 2022).

Despite all complexity of the tumour formation, one of the central events are mutations in the proto-oncogenes (e.g., *RAS*, *MYC*) that become oncogenes and in the tumour suppressor genes (e.g., *P53*, *RB*, *APC*) that normally defence cell from oncogenesis. These genes are relatively not numerous, but their impact is crucial for the malignancy manifestation. (Weinberg, 1994)

Proto-oncogenes are key factors for the regulation of the biological processes such as cell differentiation and proliferation. There are two types of the consequences of proto-oncogenes physical mutations: altered structure of the encoded protein and deregulation of the protein expression. The defective products of the oncogenes lead to uncontrolled cell division acting within cell cycle and apoptotic signalling pathways. (Kontomanolis et al., 2020) Tumour suppressor genes are crucial for cell differentiation, growth and tumour protection. For cancer proliferation, both copies of the tumour suppressor genes should be inactivated. (Kontomanolis et al., 2020)

In addition to shared traits and mechanisms, cancer cells exhibit unique properties that contribute to their distinctiveness across various types. The phenomena of tissue specificity and heterogeneity within a single cancer type are central challenges in cancer biology and therapy. (Kevin et al., 2019)

Specificity of different cancer types together with the heterogeneity of most of them posed a question about the driver mutations. There is an interest to the core mutations that launch the tumorigenesis due to the promises of targeted personalised therapy in case of this knowledge. (Kevin et al., 2019; Li et al., 2020; Alexandrov et al., 2023)

The specificity of cancer is related to the tissue-specific driver mutations. There are only few driver mutations like TERT and TP53 show broad involvement across the tissues. The understanding of the mechanisms behind the tissue specificity could unlock potential therapeutic vulnerabilities unique to different cancer types. (Kevin et al., 2019)

Several large-scale projects, like Pan-Cancer Analysis of Whole Genomes (PCAWG) Consortium of the International Cancer Genome Consortium (ICGC) and The Cancer Genome Atlas (TCGA), were consecrated to search of somatic structural variations in human cancer genome which can be potential driver mutations for different specifically for several cancer types and create characteristic mutational signature. (Li et al., 2020; Alexandrov et al., 2023; ICGC/TCGA Pan-Cancer Analysis of Whole Genomes Consortium, 2020)

If driver mutations account for tumour specificity, cancer progression imparts spatial and temporal heterogeneity, posing a significant challenge to treatment effectiveness and leading to relapse. (Kevin et al., 2019; Dagogo-Jack, 2018)

The pathological processes that sustain cancer cell viability simultaneously create molecular vulnerabilities. These vulnerabilities, often referred to as "Achilles' Heels", are not solely determined by individual gene mutations but also arise from the complex interactions between genes within tumour cells, known as gene dependency relationships (GDRs). (Howard et al., 2016; Wappett et al., 2021)

When one gene is mutated cell needs another gene to fulfil essential cellular functions. Cancer cells become reliant on specific genes or pathways to compensate for the disruption of normal cellular processes. These newly engaged genes are what is called dependency. The dependencies are interlinked through gene dependency relationships and forms a network of interconnected dependencies or GDRs. (Howard et al., 2016; Wappett et al., 2021; Pacini et al., 2024)

Thus, cancer gene mutations lead to the development of new pathological pathways that, in turn, create the need for and result in new transcriptional regulation. Therefore, GDRs are closely linked to network regulatory motifs (NRMs) of gene transcription. GDRs can be visualized and understood through network regulatory motifs. NRMs are recurring patterns in the regulatory networks that govern gene expression and cellular processes. These motifs are crucial in understanding how gene dependencies are established because they often reflect the functional interactions among genes, illustrating how certain genes work together within a network. The behaviour of NRMs helps to define the specific types of gene dependencies present within the cancer cells. (Alon, 2007; Gao, 2009; Carson, 2015; Mercatelli, 2020)

Thus, in cancer cells pathological GDR of the mutated genes can be found through network regulatory motifs of transcription. Understanding these relationships helps in mapping out the genetic vulnerabilities of cancer cells. That is crucial for advancing cancer therapies, as targeting these dependencies can exploit the molecular vulnerabilities created by the underlying genetic alterations. (Carson, 2015; Bushweller, 2019; Tao, 2023)

There is an abundant number of gene dependencies, and several research were undertaken to classify these dependencies to create a cancer dependency map which aids in the identification and ranking of potential therapeutic targets (Tsherniak et al., 2017). Two genetic screens approaches were mainly employed during these attempts, the RNAi (double-stranded RNA-mediated interference) and the CRISPR screens methods (Tsherniak et al., 2017; Pacini et al., 2024). They also called loss-of-function (LoF) screens and were used in several large-scale studies (Pacini et al., 2024). Basically, the difference between these two techniques is the target molecule and the level of effect: RNAi screens target RNA and works by post-transcriptional gene silencing; CRISPR/Cas screens target DNA, which is the genome itself, leading to DNA double-strand breaks. Also, they have different guiding agents. Small interfering RNAs (siRNAs) or short hairpin RNAs (shRNAs) serve as the guiding molecules in the RNAi mechanism to lead RNA-induced silencing complex (RISC) to the target mRNA through base-pair complementarity to degrade it. In CRISPR technology, gRNAs direct the Cas9 enzyme to specific genomic locations where it creates double-stranded breaks, thereby knocking out or modifying genes. (Yamamoto-Hino, 2013; Diehl, 2014; Yang, 2023; Setton et al., 2021)

RNAi is a straightforward and prompt method for silencing gene expression through the degradation of RNA into small fragments, which activate ribonucleases to target and degrade complementary mRNA (Agrawal, 2003). Together with the whole genome sequencing RNAi screening technology became able to be applied on the genome-wide level (Yamamoto-Hino, 2013). For studying gene transcription in mammalian, including human, cells small interfering RNAs (siRNAs) and short hairpin RNAs (shRNAs) are applicable. Additionally, the siRNAs and shRNAs libraries were developed. (Silva et al., 2004) During the gene silencing experiments the expression of targeted genes is specifically reduced (Yamamoto-Hino, 2013). RNAi loss-of-function screens are effective in pinpointing genes responsible for cellular behaviours, they can be customized to incorporate different genetic backgrounds or altered environmental settings. This screening method help uncover genetic dependencies and can be leveraged to identify potential genetic targets for therapeutic intervention. (Diehl, 2014)

Another LoF screen approach is CRISPR screens, genome-wide genetic screens which can be used across variety of cell types and are more effective and more highly applied compared to RNAi screens. RNAi screens have off-target effect and could not be used for silencing RNAs located in nucleus. CRISPR screens are highly precise and have access to all sequenced DNA. CRISPR screens are perfect tool to search of functional characteristics of the annotated genome. (Qi et al., 2017) Due to its simplicity, easy modification, and high efficiency, the CRISPR/Cas system has rapidly evolved into a versatile genome-editing tool widely used in cancer research for screening, diagnosis, and treatment. Its ability to detect genetic elements with high sensitivity makes it invaluable for rapidly identifying genes, mutations, and single nucleotide polymorphisms (SNPs) associated with tumorigenesis and cancer. (Yang, 2023) The sgRNA libraries are constructed for CRISPR genetic screens mutagenesis in mammalian cells to achieve gene inactivation. CRISPR

screens show higher level of depletion and the targetability of the protein region. (Qi et al., 2017) The CRISPR/Cas9 system is widely used for genome-wide screening of cancer-associated genes and identifying anti-cancer drug targets by utilizing sgRNA libraries to silence genes. This targeted silencing reduces cancer cell viability and helps identify potential therapeutic targets. Beyond cell line studies, CRISPR screens are also applied in animal models, offering more realistic insights into cancer biology and treatment effectiveness compared to cell line experiments alone. (Yang, 2023)

Both methods of LoF induction are used for studying synthetic lethality (SL) phenomenon, which is one of the most important variants of GDRs in cancer cells in terms of targeted therapy prospects. (Wappett et al., 2021; Doherty, 2019; Tsherniak et al., 2017; Diehl, 2014; Setton et al., 2021) SL is one of the most searched potentially efficiently targeted GDR for anti-cancer therapy with archetypal example in BRCA1/2-deficient breast cancer treatment – inhibitors of the poly (ADP-ribose) polymerase (PARP). They present the result of the first-generation clinical trials aimed at testing SL drugs. Successive rounds of focused and large-scale CRISPR screens are significantly increasing the number of SL candidate targets, many of which are in drug development. Synthetic lethality (SL) offers a framework for targeting genes that are traditionally considered "undruggable," including loss-of-function mutations in tumour suppressor genes that are essential for the development of cancer. Context-specific genetic dependencies identified through pan-cancer loss-of-function genomic screens have been found to be significantly enriched for synthetic lethality interactions. This approach enables the identification of novel therapeutic targets that can be exploited to treat tumours with specific genetic vulnerabilities. Through several large-scale project comprehensive database containing potential novel drug targets and biomarkers were created. (Setton et al., 2021) The data obtained from such large-scale studies as Project Achilles (Aguirre et al., 2016), Project DRIVE (McDonald et al., 2017), and Project Score (Behan et al., 2019; Dempster et al., 2019) was contributed to the Cancer Dependency Map (DepMap) (Tsherniak et al., 2017). (Setton et al., 2021)

SL interaction occurs when the loss of function in two or more genes leads to cell death, but the cell remains viable with a LoF in a single gene. Thus, during the co-expression of genes involved in synthetic lethality GDR cell depletion happens because of inability of a cell to survive. (Wappett et al., 2021; Setton et al., 2021)

Some studies show interesting and promising results about association of the SL with various aspects of cancer pathology. For example, the role of SL in cancer risk across different human tissues, where higher co-inactivation of cancer synthetic lethal gene pairs in normal tissues correlates with lower lifetime cancer risk and delayed cancer onset (Cheng, 2021).

Another research describes CRISPR screen application in a study of gene paralogues. The interest is that gene paralogues is that they can be potentially used to define synthetic lethal interactions. Together with CRISPR technology which can simultaneously target multiple genes across different cell lines the fitness effects of gene paralogues were investigated. A total of 105 gene pairs were identified whose co-disruption impaired cellular fitness across multiple cell lines, with 27 pairs, including the paralogues FAM50A and FAM50B, showing a significant impact. This kind of research provide insights into genetic redundancy and the survival mechanisms of cells which also provide opportunities for further therapeutic targets in cancer treatment. (Thompson et al., 2021) Another study based on paralogue gene interactions showed that mutations in one gene

increase sensitivity to inhibition of its paralogue, a mechanism that can be exploited for targeted treatments. It also highlighted that despite being a promising therapeutic target, many of paralogue genes still need to be tested for synthetic lethality across different cancer types. Additionally, the challenge lies in translating these findings into effective drugs. (Ryan, 2023)

The concepts, findings, and approaches discussed above converge in the exploration of multiple myeloma, addressing both the pathophysiological and therapeutic aspects of the disease. Multiple myeloma (MM) ranks as the second most prevalent hematologic malignancy (van de Donk NWCJ, 2021). MM is a cancer which develops from plasma cells in the bone marrow, from postgerminal center B cells (Silberstein, 2022; Cowan et al., 2022). Normal plasma cells produce infection-fighting antibody proteins while abnormal plasma cells produce dysfunctional antibodies called M proteins, which are nonfunctional intact immunoglobulins or immunoglobulin chains. (Silberstein, 2022; Brigle, 2017)

The presence of abnormal clonal plasma cells in the bone marrow with potential uncontrolled proliferation resulting in damaging bone lesions, kidney dysfunction, anaemia, and elevated calcium levels in the blood (hypercalcemia) (Cowan et al., 2022). Thus, the main systems and organs affected during this haematological malignancy are bones, immune system, blood system and kidneys. (Silberstein, 2022; Cowan et al., 2022) In a healthy individual, antigen exposure (such as from viral or bacterial infections) triggers naive B cells to proliferate and undergo somatic hypermutation in the IgH and IgL VDJ sequences (Cowan et al., 2022). This process generates long-lived plasma cells, which settle in the bone marrow and maintain humoral immunity through sustained antibody production. (Cowan et al., 2022) In multiple myeloma, abnormal clonal plasma cells mimic these normal processes but result in excessive production of intact immunoglobulins. (Cowan, 2022) Each year, approximately 588,000 people worldwide are diagnosed with MM, a cancer primarily affecting older adults with a median age of diagnosis around 69 years. While its exact cause remains unknown, risk factors include being male, obesity, and exposure to dioxins or Agent Orange. (Silberstein, 2022)

Diagnostics includes blood test (the serum protein electrophoresis and the serum free light chain assay), urine test (urine protein electrophoresis) which can detect M proteins produced by myeloma cells. Additionally, for all patients with suspicion on MM a bone marrow biopsy is recommended. Genetic and molecular testing is helpful in predicting response to treatment and estimate survival. Imaging techniques like magnetic resonance imaging (MRI), computed tomography (CT), and positron emission tomography (PET) scans are used to assess bone damage and detect plasma cell tumours (plasmacytomas) outside the bone marrow. (Silberstein, 2022)

While MM remains incurable, treatment can significantly prolong survival. In a study of 1,000 patients treated with the combination of lenalidomide, bortezomib, and dexamethasone, the median overall survival reached approximately 10.5 years. (Silberstein, 2022)

In MM patients, genetic analysis of malignant plasma cells can be performed using Fluorescence In Situ Hybridization (FISH), where fluorescent probes bind to specific chromosomal abnormalities, helping identify genetic risk factors and stratify patients accordingly. (Cowan et al., 2022)

To FISH findings associated with worse survival in MM belong 1q+, t(4;14), t(14;16), t(14;20), and del(17p) (TP53 mutation). Additional risk is conferred in case of the presence of multiple high-risk features such as the combination of del(17p) and 1q+. These mutations correspond to

following events: the t(4;14) change results in overexpression of the fibroblast growth factor receptor 3 (FGFR3) gene; del(17p) results from the deletion of the p53 gene; the t(14;16) translocation leads to the overexpression of the c-MAF oncogene. (Cowan et al., 2022)

In relapsed multiple myeloma patients, emerging FISH abnormalities suggest clonal evolution, characterized by the accumulation of additional mutations driving disease progression. The presence of a secondary 17p deletion, biallelic inactivation of tumour suppressor genes, or both, is linked to poorer prognoses. (Cowan et al., 2022)

Additional chromosomal abnormalities may include hyperdiploidy, characterized by trisomy of chromosomes 3, 5, 7, 9, 11, 15, 19, and/or 21, or the translocation t(11;14), which occurs in 15% to 20% of patients and leads to cyclin-D1 overexpression. However, these changes are not linked to a worse prognosis. (Cowan et al., 2022)

The primary goal of treating multiple myeloma is to enhance both survival and quality of life by managing disease-related complications and maintaining long-term control over the malignancy. Treatment strategies focus on reducing the population of malignant plasma cells in the bone marrow, which is partly monitored by levels of monoclonal proteins and serum free light chains. A greater reduction in these malignant cells is associated with more sustained disease management. (Cowan et al., 2022)

Significant advancements in multiple myeloma treatment have come from the development of proteasome inhibitors (such as bortezomib, ixazomib, and carfilzomib), immunomodulatory agents (including thalidomide, lenalidomide, and pomalidomide), monoclonal antibodies targeting myeloma cell surface antigens (like daratumumab, elotuzumab, and isatuximab), and the use of autologous hematopoietic stem cell transplantation. (Cowan et al., 2022)

Multiple myeloma (MM) poses challenges due to drug resistance and relapse. While patients may initially respond to therapy, resistance often develops, leading to treatment failure. Contributing factors include genetic and epigenetic changes, altered drug metabolism, reduced intracellular drug levels, dysregulation of apoptosis and signalling pathways, activated autophagy, and the persistence of drug-resistant cancer stem cells. Additionally, MM cells' reliance on the stromal microenvironment plays a role. Treatment itself can exert selection pressure, fostering the survival of resistant cancer cell populations. (Das, 2022)

The molecular concept of drug resistance in MM is based on the P-glycoprotein (P-gp) functioning. P-gp, produced by the MDR/ABC B1 gene, plays a key role in drug resistance, particularly in treatments involving proteasome inhibitors, alkylating agents, and immunomodulatory drugs. As part of the pump transporter family, P-gp mediates the efflux of drugs from cells. About 75% of patients treated with drugs like doxorubicin, dexamethasone, and vincristine exhibit P-gp expression, whereas untreated MM patients show lower levels, indicating cumulative dosing increases P-gp expression. Melphalan treatment does not significantly increase P-gp levels, but other drugs like vincristine and bortezomib do. P-gp inhibition is a target in clinical studies, and it may serve as a predictive marker for therapeutic response in MM. Multidrug resistance protein (MRP), along with other ABC family members like BCRP and LRP, also contributes to drug resistance, particularly with drugs like doxorubicin, daunorubicin, and vincristine. Multidrug

resistance arises when cancer cells resist multiple, unrelated drugs after exposure to one chemotherapeutic agent. (Das, 2022)

During the treatment of the MM relapses occur at various stages of the disease due to its biological heterogeneity. "Relapsed myeloma" refers to previously treated myeloma that has returned or progressed following earlier therapy, necessitating a new course of treatment. Treatment schemes and protocols of the MM are complex and include several types of anti-cancer drug classes, such as immunomodulatory drugs, proteasome inhibitors, monoclonal antibodies, chemotherapy, some specific drugs (venetoclax, selinexor). They are often administered in combination, however, each of them has its proper relapse mechanism. (Bhatt, 2023)

Protein inhibitors (PI) have their pharmacological action primarily due to their suppression of NF- κ B signalling, affecting both canonical and non-canonical pathways by inhibiting I κ B and p100/105 proteins. This reduction in NF- κ B activity induces anti-myeloma and pro-apoptotic effects by disrupting the regulation of caspase inhibitors, Bcl-2 proteins, mitochondrial cytochrome c release, and cytokine production. However, the key anti-myeloma effect arises from the build-up of misfolded proteins, normally degraded by the 20S proteasome, which triggers apoptosis through endoplasmic reticulum stress, known as the terminal unfolded protein response. (Bhatt, 2023)

Nevertheless, proteasome inhibition has emerged as a crucial therapeutic strategy in the treatment of MM, because of the role of the ubiquitin proteasome system in tumorigenesis including myeloma oncogenesis (Ito, 2020; Zhou, 2023).

Proteasome-encoding genes have been found to exhibit elevated expression in various tumours, suggesting they may serve as promising new targets for cancer treatment. This overexpression highlights the potential of proteasome inhibitors as therapeutic agents in oncology. (Zhou, 2023)

The ubiquitin-proteasome system (UPS) plays a crucial role in protein degradation and cellular regulation controlling quantity and quality of the proteins in the cell. UPS influences various biological processes, including cell cycle progression, apoptosis, and DNA repair. This system tags proteins with ubiquitin, marking them for degradation by the proteasome. (Hanna, 2019)

In the context of cancer, particularly MM, the UPS is significant for its role in regulating the levels of oncoproteins and tumour suppressors. Proteasome inhibitors, such as bortezomib, have been developed to disrupt this system, leading to the accumulation of pro-apoptotic factors and the induction of apoptosis in malignant cells. This is particularly effective in MM, where the overexpression of proteasome-encoding genes has been linked to tumour growth and drug resistance. PS-341 (bortezomib) inhibits growth, induces apoptosis, and overcomes drug resistance in multiple myeloma cells by binding to the 26S proteasome. This binding blocks the degradation of ubiquitinated proteins, leading to the accumulation of key regulatory proteins such as NF- κ B, p53, and cyclins, which promotes apoptosis. Furthermore, the buildup of misfolded proteins triggers cell death through the unfolded protein response, making proteasome inhibitors an effective strategy for targeting multiple myeloma. (Cockram et al., 2021; Devasia, 2023)

Because of the involvement of the SL processes in MM pathology (Teoh et al., 2022; Zhang et al., 2023) and effective treatment result of the proteasome inhibitors (Ito, 2020), it would be interesting to search for GDR of the SL type with engagement of genes responsible for ubiquitin-proteasome system (UPS) system functioning. SL application in MM can be represented by two examples. First, NEIL1-p53 abnormalities with Chk1 activation have SL relationships and are possible Achilles heels (Teoh et al., 2022). Second, histone deacetylase 1 and 2 (HDAC1 and HDAC2) paralogues which are synthetically lethal can cause nucleosome remodelling and deacetylase (NuRD) complex partly damage that is MM vulnerability (Zhang et al., 2023). Importance of UPS is reflected in MM treatment strategies, where proteasome inhibitors are a crucial part for over a decade, with bortezomib being the first approved PI by the FDA in 2003 (Ito, 2020).

However, only one paper of 2011 was found that precisely describes a possible junction point between these three phenomena: MM, SL and UPS (Neri et al., 2011). It discusses the mechanism when proteasome inhibitor bortezomib disrupting DNA repair process in MM cells is combined with PARP inhibitors and this resulting in more effective cancer cells destruction than either factor alone (Neri et al., 2011).

Additionally, despite recent advancements in treatment, multiple myeloma remains largely an incurable cancer. Focusing on tumour-specific essential genes that represent druggable dependencies may provide a promising approach for developing new therapeutic agents to combat MM and address drug resistance (Went, 2022).

This data suggests that gaining deeper insights into both GDRs and their associated molecular vulnerabilities in MM, as well as the role of UPS in this haematological malignancy, could offer valuable perspectives for identifying new therapeutic targets.

The actual study presents an attempt to approach in understanding of NRM that underly the GDR through gene pairs dataset containing transcriptional information from samples of MM patients sourced from Multiple Myeloma Research Foundation (MMRF) (Shaw, 2021; Settino, 2021) site. Before the present analysis it was treated by MultiSEp method (Wappett et al., 2021) in the Overton Laboratory by Ms Adeline McKie. One of the principled aims of analysis was to create co-expression depletion patterns for gene pairs to reveal NRM which can be useful in further understanding of GDRs between genes in the dataset. Gene pair lists and gene lists were created with possibility of their subsequent ontology enrichment and functional annotation which is also needed for deeper insight of the GDRs between gene pairs in MM samples.

The application of the MultiSEp algorithm to the precursor dataset formed the initial dataset and can be explained by the MultiSEp algorithm's principles of working. (<https://www.overton-lab.uk/app/synlegg/#tuid>; Wappett et al., 2016)

The MultiSEp package in R analyses polyomics data across numerous samples and is integrated into the SynLeGG tool for comprehensive synthetic lethality analysis of RNA-Seq data from the Cancer Cell Line Encyclopedia (CCLE). Derived from the BiSEp approach, MultiSEp employs a refined statistical method for modeling complex data distributions, both designed to analyze

synthetic lethality interactions in cancer cells. (<https://www.overton-lab.uk/app/synlegg/#tuid>; Wappett et al., 2016)

Synthetic lethality occurs when simultaneous loss of function (LoF) in two or more genes causes cell death, while cells survive if these LoF events occur individually. A LoF mutation is a genetic alteration that reduces or eliminates gene function, disrupting cellular processes. Such mutually exclusive losses imply that one gene compensates for the other's loss, making their combined LoF lethal, presenting a potential synthetic lethality scenario. (<https://www.overton-lab.uk/app/synlegg/#tuid>; Wappett et al., 2021)

Thus, this work represents an effort to make a small step forward in understanding the mechanisms of multiple myeloma by establishing a foundation for exploring gene dependencies that could lead to potential drug targets.

BiSep identifies mutually exclusive LoF events in gene pairs from multiomics cancer datasets, integrating gene expression data. It detects cases where two genes are rarely mutated together. Based on the Bimodality principle, it calculates the Bimodality Index (BI) using the BIG (Bimodality in Genomics) algorithm to identify bimodal gene expression patterns and detect functional deficiencies. (Wappett et al., 2016)

Additional tools include BiGEE (Bimodal-low Gene Expression Exclusivity) and BEEM (Bimodal Expression Exclusive with Mutation). BiGEE uses BiSep's log₂ expression matrix to find gene pairs with mutually exclusive low expression, identifying potential synthetic lethal pairs. BEEM links gene expression with mutation profiles, prioritizing mutations enriched in high expression states. (Wappett et al., 2016)

MEMU (Mutually Exclusive Mutations) module identifies gene pairs with mutually exclusive mutations, while FuRE (Functional REDundancy between synthetic lethal genes) module ranks functionally redundant gene pairs based on their semantic similarity from MEMU, BiGEE, or BEEM outputs. (Wappett et al., 2016)

Statistical tests, including Fisher's exact test and corrections for multiple comparisons (e.g., Bonferroni or FDR), ensure reliable results, highlighting gene pairs with the most significant findings. (Wappett et al., 2016)

MultiSep analyses not only genomic, but additionally RNA-seq, CRISPR, and mutational data, to enhance the accuracy of its predictions. This polyomics approach is crucial for understanding the complex interactions in cancer biology. MultiSep is designed to identify GDRs by clustering gene expression data from various cancer cell lines. It helps in discovering potential therapeutic targets by analyzing how the loss of one gene can make a cell vulnerable to the inhibition of another gene, a concept known as "Achilles' heels" in cancer biology. (Wappett et al., 2021; <https://www.overton-lab.uk/app/synlegg/#tuid>)

The performance of MultiSep was evaluated against other methods, such as BiSep, using synthetic lethal gene pairs from the SynLethDB database. MultiSep provided a much larger coverage of candidate gene pairs, being 5.8-fold higher than BiSep. This extensive coverage allows for a more comprehensive analysis of genetic interactions. (Wappett et al., 2021)

BiSEp limits its analysis by partitioning cell lines into only two groups per gene. This binary approach can overlook subtler dependencies that may exist within the data, as it only considers bimodal distributions that are statistically significant. MultiSEp employs a distinct clustering method that sets it apart from BiSEp in the analysis of genetic dependencies. (Wappett et al., 2021) The advantage of the MultiSEp algorithm is that it can analyse the expression relationships of two genes across more than two clusters, generating between two and five clusters of sample types for each gene. Unimodal models are not considered, as a single cluster would conflict with the partitioning necessary for downstream analysis. This provides deeper dissection of synthetic lethality phenomenon and enables to discover other types of GDRs. MultiSEp assigns each cell line a probability of belonging to each cluster, which determines its final cluster assignment, especially in cases where a cell line falls within an overlapping region between two clusters. That is possible due to the application of Gaussian mixture modelling (GMM) with Expectation-Maximization to identify clusters of gene expression, with the number of clusters being determined through Bayesian Information Criterion (BIC) regularization. The GMM analysis addresses the limitations of the BiSEp approach, which restricts cell lines to two groups per gene and only allows comparisons when bimodality is statistically confirmed. (Wappett et al., 2021; Lubbock et al., 2013; <https://www.overton-lab.uk/app/synlegg/#tuid>)

This approach was first used in TMA Navigator tool for studying tissue microarrays (TMAs) to perform simultaneous analysis of multiple tissue samples with the aim to evaluate biomarker protein expression in tumour biopsies. (Lubbock et al., 2013)

The GMM is employed to identify clusters of samples based on their marker expression values. This method provides significant advantages applying statistically sound and automated identification of expression groups. It accommodates the inherent complexity of gene expression profiles. The GMM fits a mixture of Gaussian distributions to the marker values using an expectation-maximization algorithm. Each cluster is characterized by its own mean and standard deviation, which allows for a more biologically relevant representation of the data compared to fixed standard deviations across clusters. The GMM allows for unsupervised grouping, meaning that it does not require prior knowledge of the groups within the data. Instead, it automatically determines the number of clusters present in the dataset. The number of clusters, or cardinality, is selected using BIC regularization. BIC is a statistical criterion that helps in model selection by balancing model fit and complexity. It penalizes models with more parameters, thus favouring simpler models that adequately explain the data. (Lubbock et al., 2013; Wappett et al., 2021; <https://www.overton-lab.uk/app/synlegg/#tuid>)

MultiSEp is a key algorithm utilized in the SynLeGG framework for analysing and visualizing multi-omics data, particularly in the context of cancer research. The results of the GMM can be visualized through density plots and histograms, which help in understanding the distribution of marker scores across the identified clusters. This visualization aids in interpreting the biological significance of the clusters. The groups identified through GMM can be further analysed using survival analysis techniques, such as Kaplan-Meier plots. This integration permits to assess the

prognostic implications of the identified clusters based on marker expression. (Wappett et al., 2021)

This dissertation consists of the follow parts:

- 1) Abstract summarising the content of all sections;
- 2) Introduction with background of knowledge related to the topic and dissertation overview;
- 3) Materials and methods chapter with description of the technical basis represented by hardware, software, structure and biological basis of the dataset; it also contains detailed description of the dataset analysis steps; explanation of the MultiSEp and statistical approaches preceding the actual analysis of the dataset;
- 4) Results chapter containing the outcomes appeared on each step of the dataset analysis including pattern analysis and gene pairs and single gene lists generation;
- 5) Discussion chapter which includes interpretation of the principal observations: cluster counts, revealed patterns, quantitative gene lists analysis; consequences and implications of the outcomes;
- 6) Conclusion chapter with the summary of the key findings, recommendations and final thoughts and words;
- 7) Future work and limitations chapter with discussion of the limitations and further research suggestions.
- 8) List of References and Appendices in the end.

II. MATERIALS AND METHODS

To perform computational analysis, computing hardware with appropriate technical characteristics is required, such as a CPU (Central Processing Unit), RAM (Random Access Memory), a storage system (SSD vs HDD), a GPU (Graphics Processing Unit), support for parallel computing and clusters, and high I/O performance (input/output performance). Due to the large size of the dataset file, an HPC (High-Performance Computing) cluster and local machines had to be used for data storage to provide the necessary memory capacities for processing. The operating system is important. The most convenient for bioinformatics is Linux OS, especially when terminal is used for programming purposes and dataset analysis.

The dataset itself is the central part of the research, where structure and biological background are significant.

The third component is a software, program, computational tool or programming language that is employed for analysis. It can be related to both materials and methods and reflects the principles of the dataset analysis.

2.1 COMPUTING HARDWARE

The storage of the dataset and of the files produced during the analysis was supplied by Kelvin2 High Performance Computing (HPC) and Research Data Storage environment. The common Kelvin2 HPC characteristics were: 96 x 128 core Dell PowerEdge R6525 compute nodes with AMD EPYC 7702 dual 64-Core Processors (786GB RAM); 8 High memory nodes (2TB RAM); 32 x NVIDIA Tesla v100 GPUs in 8 nodes; 16 x NVIDIA Tesla A100 GPUs in 4 nodes; 2PB of lustre parallel file system for scratch storage; Compute nodes run CENTOS7 operating system. (<https://www.ni-hpc.ac.uk/Kelvin2/>)

The local kelvin2 cluster has the following characteristics: architecture x86_64; CPU op-mode(s) were 32-bit, 64-bit; CPU(s) number was 6; CPU model name: QEMU Virtual CPU version 1.5.3; CPU clock speed was 2595.122 MHz. The Kelvin version was r2024.1 based on Rocky Linux 8.10.

There was a filesystem on the Kelvin2 cluster with the file distribution system protocol which was NFS type with 6.0P size. This file system had a main storage directory with subdirectory storing laboratory group disk space ('NetBiology'). 'NetBiology' folder contained several folders for each laboratory user. In one of these folders the file with the initial dataset of the present analysis was kept. The initial file had a .txt extension.

There were two nodes used during the present study: an operating node (personal computer) where all analysis was implemented and a storage node (local storage machine) where the files needed for analysis and the files issued from analysis were stored.

The file containing the initial dataset was copied into the folder of the local storage machine called knappogue, which was then used for the storage of the obtained resulting files. Kelvin2 was further used for backup of the data.

The knappogue local machine had the following characteristics: architecture x86_64; CPU op-mode(s) 32-bit, 64-bit; CPU(s) number was 24; model name was Intel(R) Xeon(R) CPU E5-2620 v3 @ 2.40GHz; CPU maximal clock speed was 3200.0000 MHz and minimal clock speed was 1200.0000 MHz. The filesystem used on the knappogue machine was /dev/sdc1 with 1.9T size. The actual size of the disk of the working directory was 126 G.

Files were stored on the knappogue in two folders because of the voluminous files' size.

The local personal computer node had the following characteristics: architecture x86_64; CPU op-mode(s) 32-bit, 64-bit; CPU(s) number was 8; model name was Intel(R) Core(TM) i7-3770S CPU @ 3.10GHz; CPU maximal clock speed was 3900.0000 MHz and minimal clock speed was 1600.0000 MHz.

2.2 SOFTWARE: DATA ANALYSIS AND VISUALISATION TOOLS

Because the initial file and several generated from it files were bulky, the Linux operation system terminal was used for analysis procedure. The main part of the analysis was conducted in R using the *dplyr*, *tidyverse*, and *stringr* packages.

The Linux operating system was presented by the openSUSE Leap 15.5 version of the openSUSE Linux distribution. The analysis was held in the terminal: platform x86_64-suse-linux-gnu; arch x86_64; os linux-gnu; system x86_64, linux-gnu.

R program characteristics were R version 4.1.2 (2021-11-01). R program was running using the Linux terminal.

PowerPoint and Excel programs were used for the consolidation of principal results and for the visualization of the results. PowerPoint and Excel characteristics: version 2108 built on Microsoft® PowerPoint® LTSC MSO (16.0.14332.20761) 32-bit.

2.3 DATASET

2.3.1 Preliminary preparation of the dataset for the present study

A precursor dataset was obtained from the MMRF (Multiple Myeloma Research Foundation) CoMMpass (Clinical Outcomes in Multiple Myeloma to Personal Assessment) data resource (<https://themmr.org/>; Shaw, 2021; Settino, 2021), IA21 data section, “Expression Estimates – Gene Based” rubric: ‘MMRF_CoMMpass_IA21_salmon_geneUnstranded_tpm.tsv’. It was processed using the MultiSEp package in R (Analysis was run in the Overton Laboratory by Ms Adeline McKie). Using the MultiSEp algorithm, RNA-seq data can be statistically processed, decomposed, and restructured to identify cell clusters formed by different gene

expressions. (Analysis was run in the Overton Laboratory by Ms Adeline McKie) (Wappett et al., 2021; <https://www.overton-lab.uk/app/synlegg/#tuid>)

The dataset for present analysis was called Myeloma_XPR_GDR dataset. It contained 928 patients. It was preliminary processed in MultiSep R package and statistically analysed by Ms Adeline McKie. The dataset resulting from the MultiSep analysis and used for the present study includes only the cluster combinations formed by statistically significant gene pairs' expression. This significance was tested during the formation of the initial dataset. Primarily, the chi-square association test was applied for the whole contingency table issued from MultiSep analysis to prove the hypothesis that the association between all cells in that table is significant. In other words, to test whether the distribution of data points across clusters is dependent. (Analysis was run in the Overton Laboratory by Ms Adeline McKie) (Wappett et al., 2021; <https://www.overton-lab.uk/app/synlegg/#tuid>)

Subsequent Binominal test for each cell of the contingency table was then executed. The statistical significance of gene expression for each cell table was assessed by comparing the expected sample count issued from MultiSep prediction with the actual sample count derived from biomedical data. This comparison is carried out through hypothesis testing whether the observed distribution of sample counts in gene expression clusters significantly deviates from what would be expected under a null hypothesis of independence or no mutual exclusivity between gene pairs. That provides evidence for potential synthetic lethality or functional interaction between the genes in question. (Analysis was run in the Overton Laboratory by Ms Adeline McKie) (Wappett et al., 2021; <https://www.overton-lab.uk/app/synlegg/#tuid>)

The mutual exclusivity within gene pairs is tested by Binomial test, the result of which is presented in the dataset in form of “p-value” column. (Analysis was run in the Overton Laboratory by Ms Adeline McKie)

Due to the substantial number of observations in the dataset, indicating a high statistical power, calculating the q-value is necessary to control the false discovery rate during multiple hypothesis testing. One of the most effective ways to treat this issue called multiplicity problem is the Benjamini-Hochberg procedure. This method focuses on the false discovery rate (FDR), which is defined as the expected proportion of false positives (incorrect rejections of the null hypothesis) among the rejected hypotheses. (Benjamini, 1995)

Controlling the FDR is particularly beneficial in cases where many hypotheses are tested simultaneously, as it helps to make more discoveries without excessively increasing the risk of false positives. The Bonferroni-type procedure controls the FDR for independent test statistics which enhances statistical power and provides more significant research findings. (Benjamini, 1995)

The q-value is a specific measure derived from FDR. It represents the minimum FDR at which a particular hypothesis test may be considered significant. The q-value provides a way to adjust p-values for multiple comparisons so that the FDR can be controlled while making decisions about which hypotheses to reject. (Benjamini, 1995)

The q-value produced for each gene pair can be found in the “New q-value” column. (Analysis was run in the Overton Laboratory by Ms Adeline McKie)

The initial dataset contains only the gene pairs with significant p-values and q-values, which were selected from the precursor dataset after MultiSEp analysis. (Analysis was run in the Overton Laboratory by Ms Adeline McKie)

In other words, the dataset generates quantified and statistically validated data about gene expression ratios for different gene pairs (Gene1 and Gene2) in multiple myeloma samples. Hereby, this provides an opportunity for further gene dependency relationship (GDR) analysis by studying the distribution patterns of cluster combinations within gene pairs.

2.3.2 Dataset structure

The Myeloma_XPR_GDR dataset represents itself as a table which consists of the 12 following columns: Gene1, Gene2, Gene1_ClusterCount, Gene2_ClusterCount, Actual_Count, Expected_Count, TotalCluster_Combinations, Cluster_Combination, Patients_Count, p_value, q_value, New_q_value. Examples of the data in each column are presented in Table 1.

The Myeloma_XPR_GDR dataset contains information about the samples clustering depending on the pair of genes (Gene1 and Gene2) expression ratio. The expression level of each studied gene contributes to the sample dissemination on the graph and forms its own clusters. Then, the merging of individual genes' clusters forms cluster combinations. These cluster combinations together create the total cluster combination view which is put into grid representing the gene pair functional interaction result. This grid appeared after MultiSEp pairwise analysis of both genes' clusters based on the transcriptomics data in the precursor dataset. It is essentially a contingency table mentioned earlier, which was then statistically tested by chi-square test to prove an association between cluster combinations within one table followed by Binomial test for each cluster combination. However, the Myeloma_XPR_GDR dataset contains information only on the Binomial test results, which are shown in the p-value column and by interaction in New_q-value column for multiple testing correction of these p-values. Thus, the Myeloma_XPR_GDR dataset conveys information about pairs of genes' co-expression clustering of multiple myeloma samples. The table cell corresponds to the cluster combination. The number of cluster combinations determines the number of table cells. The total number of cluster combinations for each gene pair is different and determines table size. It affects also the table form, because each gene from gene pair can invest different number of clusters in the total cluster combination (table). The minimum number of clusters from each gene to create a table is 2, and the maximum is 5. Thus, the smallest table is 2x2 in size and consists of 4 table cells (4 cluster combinations); the biggest table is 5x5 in size and consists of 25 table cells (25 cluster combinations). The table form has two variants in case of the different number of clusters appearing from expression of each gene in gene pair. For example, it can be 3x5 size which gives vertical form, or 5x3 which gives horizontal form. The relation between table form and orientation will be explained further in the text.

The term “table type” means in further text primarily the table size which is total number of table cells, or total cluster combination, and additionally the table form.

Generally, all types of tables are possible in the dataset between 2x2 and 5x5 table dimensions with the statement that one of each side or both sides consist of 2, 3, 4, or 5 cells. Because the comparison of at least two gaussians happens to create the contingency table, it is impossible for table to have less than 2 cell in both horizontal and vertical directions. Thus, it is impossible to have a number of cells equal to primary number like 11 or 17.

Here, the x-axis and the y-axis notions need to be introduced, which provides the coordinates for each cluster combination inside the table. Horizontal tables have more cluster combinations by their x-axis and vertical tables – by their y-axis. Oppositely, vertical tables have more cluster combinations by their y-axis side and less by the x-axis side.

The form of the table is important, because it shows which gene expression in gene pair generated more clusters in scale of the whole table. The table form also reflects the possible expression levels of Gene1 and Gene2 within each gene pair in terms of number of clusters that each of two genes can form. The information about the table form appoints further calculations.

As far as the category of a cluster combination reflects the coordinates, or position, of a cluster combination in the gene expression clustering table, it reflects the level of expression for both of two genes involved in the formation of the cluster combination. The level of expression can be identified by the n and the m meanings, or x-axis and y-axis coordinates of the cluster combination. Thus, if a cluster combination belongs to “1_x_2” category, it means that in the pair of genes which has cluster co-expression in this quadrant of a table, Gene1 has higher relative level of expression than Gene2.

A certain configuration of certain cluster combinations in a certain table type is a pattern. In other words, pattern is an assembly of cluster combinations with their individual characteristics for one table type.

Table 1. The meaning of each column of the dataset.

Column No	Name of the column	Example
1	Gene1	ENSG00000006062
2	Gene2	ENSG00000166822
3	Gene1_ClusterCount	5
4	Gene2_ClusterCount	4
5	Actual_Count	66
6	Expected_Count	117.989224137931
7	TotalCluster_Combinations	20
8	Cluster_Combination	2_x_1
9	Patients_Count	928
10	p_value	2.42915088901628e-08

11	q_value	3.27999867558768e-06
12	New_q_value	5.67951771023447e-06

Gene1 and *Gene2* columns represent gene pairs with Ensembl ID for each gene. That means that the gene names start from “ENSG” letters and 11 numbers after.

Gene1_ClusterCount and *Gene2_ClusterCount* columns contain information about the number of clusters, formed by each gene expression in the gene pair. In the pattern grid it corresponds to the number of table cells for x-axis created by *Gene1* and for y-axis created by *Gene2*. This data provides information about the table form (horizontal or vertical) and demonstrates the varying contribution of each gene to the overall expression. These two columns are related to the *TotalCluster_Combinations* column which shows the total number of cluster combinations in table which resulted from *Gene1_ClusterCount* and *Gene2_ClusterCount* columns multiplication.

Actual_Count and *Expected_Count* columns refer to information about the number of samples that each cluster combination contains. *Actual_Count* column shows the actual number of samples, taken from RNA-Seq results, in each cluster combination in a gene pair. *Expected_Count* column shows the number of samples for this cluster combination that would be expected through MultiSEp statistical prediction. (Analysis was run in the Overton Laboratory by Ms Adeline McKie) First, MultiSEp counts the number of samples in cluster combinations for each of two genes and then combines these calculations for both genes using regularization of the number of gene expression clusters. As a result, the probabilities for each of the analyzed genes from the gene pair were multiplied so that the samples’ proportion for each cluster combination in total cluster combination was calculated. (Wappett et al., 2021; <https://www.overton-lab.uk/app/synlegg/#tuid>) The principles of these statistical calculations in detail will be discussed later. This information was not used in the present research. However, it is important for further deeper analysis of found patterns’ characteristics because the number of living samples is just as important as whether they are dead or alive. It gives more profound information about the depletion type of the pattern and thus, the GDR category.

These two columns are indirectly connected to the p-value and q-value columns which contain the results of the Binomial test p-values and their multiple testing corrections as q-values.

TotalCluster_Combinations column results information from the *Gene1_ClusterCount* and *Gene2_ClusterCount* columns and represents the total number of the table cells in the whole table. There are 10 possible meanings for this column: 4, 6, 8, 9, 10, 12, 15, 16, 20, 25. Thus, *TotalCluster_Combinations* column refers to information about the table size.

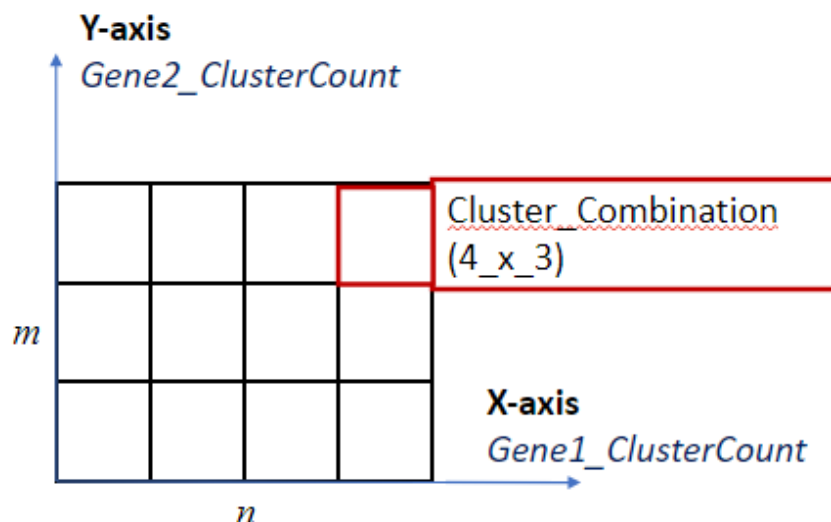


Figure 1. Explanation of the table organization. In “Cluster_Combination (4_x_3)”, $n=4$, $m=3$, TotalCluster_Combinations=12 and the table type is 12(4x3).

Cluster_Combination column contains each cluster combination coordinates regarding the cluster combination position relative to the x-axis and the y-axis. The cluster combination coordinates have a form of “ n_xm ”, where n is a cluster position by the x-axis and m is a cluster position by the y-axis. Biologically, n meaning corresponds to the number of clusters produced by Gene1 and m meaning to the number of clusters for Gene2. Thus, together they generate one cluster combination table cell with n_xm meaning. (Figure 1)

Patients_Count column contains the information about the number of patients whose samples were selected to be included in the precursor dataset, that was downloaded from MMRF CoMMpass source (<https://themmrf.org/>; Shaw, 2021; Settino, 2021). These values are retained in the Myeloma_XPR_GDR dataset (MMRF dataset that was treated by MultiSEp).

p_value column conveys information about the statistical significance of a cell in a table, or cluster combination in a total cluster combination, according to the Binomial test.

q_value column not operating column because of the technical bias.

New_q_value column reflects the multiple testing correction for the p-value results; approve that there are no false positives among p-values.

2.3.3 Additional explanations

Gene1_ClusterCount, *Gene2_ClusterCount* and *TotalCluster_Combinations* columns are technically and conceptually interconnected, because they all are responsible for the table form and type. The union of these three columns is unique for each gene pair. On the contrary, several gene pairs correspond to one type of union of these three columns.

Cluster_Combination column’s meaning can be the same for different gene pairs, because different gene pairs cluster combinations can have similar coordinates meanings; this is due to the

fact, that different table types always contain clusters with similar coordinates. For example, all table types possess cluster combination with “1_x_1” position.

The uniqueness of the cluster combination (table type and position characteristics) is mostly defined by the gene pair (*Gene1* and *Gene2* columns) and the table type. However, to one cluster combination in the certain table type correspond several gene pairs. The table type combines these two conditions and links cluster combination data with gene pair information, which together form the pattern. Thus, a pattern of gene expression for multiple gene pairs is a mixture of certain cluster combinations categories in a certain table type. This certain cluster combinations’ mixture is important because that is where the biological event of interest happens and (samples are dead or alive) is reasoned by the two genes’ co-expression. Several gene pairs can correspond to the one pattern, but each pattern is unique for a gene pair.

Though the actual sample count data was not employed in the present study, it is still important both for cluster combination and for pattern characteristics.

III. RESULTS

Three main procedures were undertaken with the dataset: cluster combinations count, patterns assemblage and gene pairs and single genes' lists analysis. They were divided into six technical steps: cluster combinations count before and after standardization of table forms, standardization, pattern assemblage, gene pairs and single genes' lists development and gene lists quantitative analysis. Before proceeding with these steps, preliminary dataset analysis was performed to identify the general structure of the data and the existing characteristics of its principal elements (table sizes, table forms, cluster combinations categories).

3.1 ANALYSIS WORKFLOW AND FILES MANAGMENT

To summarize the sequence of the analysis steps, the workflow was constructed.

The computational workflow to determine complex patterns of gene dependency relationships is a key result of the present work.

Analysis steps are related to dataset transformation stages related to analysis steps and files produced from these transformations. After receiving the standardized dataset and then pattern dataset as single files, the multiple sub-dataset files were created for more convenient data manipulation. Unique patterns were extracted and counted for each table type file. Unique genes in pairs and single were taken from divided patterns file. Common and different gene lists were extracted from single gene lists. For all table types except 25-cell type 3 prevalent patterns were taken to find gene pairs and single gene lists. Common and different genes in single gene lists between patterns in one table type was found for all table types except 25-cell type.

For clarity it was put in the form of a scheme. (Figure 2)

FILES SIZES: The complete list of resulting files, along with their sizes, can be found in the Appendix.

FILES FORMAT: All resulted from analysis files were created in .txt extension.

Full data of the resulting files can be found in the Appendix.

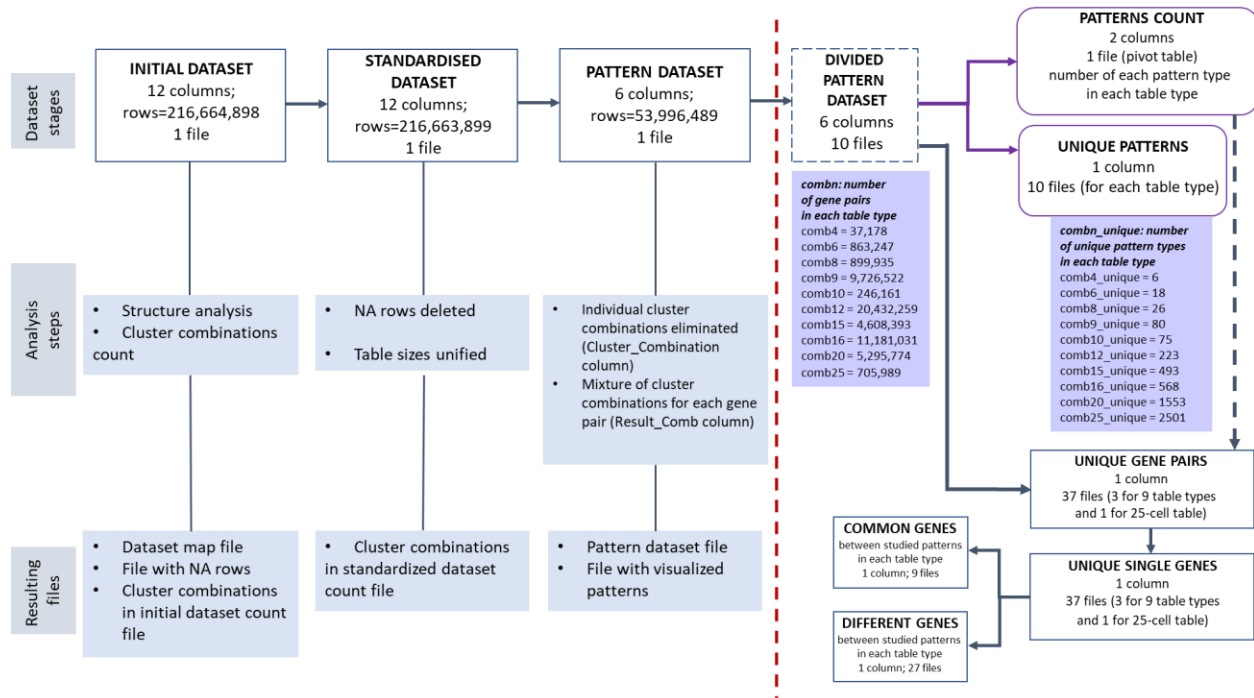


Figure 2. Overview of computational workflow. Left part of the scheme (from the red dashed line) shows the transformation stages of the whole dataset (dataset that contained all table types) from initial through standardized to the pattern dataset. For these three big datasets dataset stages, analysis steps and resulting files are indicated. There is information for each of them about the number of columns and number of rows. 12 columns that were in initial and standardized datasets are: Gene1, Gene2, Gene1_ClusterCount, Gene2_ClusterCount, Actual_Count, Expected_Count, TotalCluster_Combinations, Cluster_Combination, Patients_Count, p_value, q_value, New_q_value. 6 columns for pattern dataset are: Gene1, Gene2, Gene1_ClusterCount, Gene2_ClusterCount, TotalCluster_Combinations, Result_Comb. After the red dashed line there is a sequence of sub-dataset development stages from 10 pattern sub-datasets created from big patterns dataset to gene lists. From each of these 10 datasets two types of information portions were obtained. First portion was about patterns: a) count of each pattern type in each table type (how many gene pairs refer to each pattern type) (“PATTERN COUNT” on the scheme) and b) unique patterns’ lists (enumeration of pattern types) with number of unique patterns for each table type (what kinds of patterns are distributed between gene pairs inside table type and how many of these kinds) (“UNIQUE PATTERNS” on the scheme). In “PATTERN COUNT” one column contained each of n (from “UNIQUE PATTERNS”) types of a pattern and second column contained number of gene pairs corresponding to this pattern type (count). The second portion was about unique gene pairs for chosen patterns; these gene pairs were counted and extracted from each table type pattern sub-dataset. There is a dashed arrow between the “PATTERN COUNT” and the “UNIQUE GENE PAIRS” because the patterns to which unique gene pairs were analyzed were chosen from “PATTERN COUNT” file. From “UNIQUE GENE PAIRS” files “UNIQUE SINGLE GENES” files were created. From “UNIQUE SINGLE GENES” files common and different genes were extracted for all table types except 25-cell type. combn files

contained information about the number of gene pairs in each table type after big pattern file partition. *combn_unique* files contained information about the number of types of unique patterns in each table type. Number of columns and files created are given for all datasets and sub-datasets. Prevalent patterns visualization was based on the “PATTERN COUNT” dataset.

3.2 PRELIMINARY ANALYSIS: DATASET STRUCTURE

In the beginning to understand dataset structure the preliminary analysis of the dataset was performed. There were 216,664,898 rows in the dataset which corresponds to the number of gene pairs in the dataset. 999 rows were NA type which means that no column contained information for these rows. Thus, the actual number of analyzed gene pairs was equal to 216,663,899.

The next step was to find table types, including all table forms and table sizes present in the dataset. This was made by sorting and summarizing the data *Gene1_ClusterCount*, *Gene2_ClusterCount* and *TotalCluster_Combinations* columns. 16 types of tables were found: 4(2x2), 6(2x3), 6(3x2), 8(2x4), 8(4x2), 9(3x3), 10(2x5), 10(5x2), 12(3x4), 12(4x3), 15(3x5), 15(5x3), 16(4x4), 20(4x5), 20(5x4), 25(5x5). Thus, there were 10 types of table sizes, regarding the number of cluster combinations that they had: 4, 6, 8, 10, 12, 15, 16, 20, 25.

Among all these tables there were 25 cluster combination categories: "1_x_1", "1_x_2", "1_x_3", "1_x_4", "1_x_5", "2_x_1", "2_x_2", "2_x_3", "2_x_4", "2_x_5", "3_x_1", "3_x_2", "3_x_3", "3_x_4", "3_x_5", "4_x_1", "4_x_2", "4_x_3", "4_x_4", "4_x_5", "5_x_1", "5_x_2", "5_x_3", "5_x_4", "5_x_5".

The pivot table for the results about table types and the number of gene pairs corresponding for these table types was created to take in evidence the scale of each table type and the repartition of gene pairs count between them (Table 2). According to these results, the predominant number of the gene pairs belong to the following table types: 9(3x3), 12(3x4), 12(4x3), 16(4x4). Table types 15(3x5), 20(4x5) and 20(5x4) have a more important number of gene pairs forming them compared to the other table types as well. To summarize, table type 12(3x4) is formed by the highest number of gene pairs expression which is 46,729,054; table type 16(4x4) is close to the prevalent table type and is formed by 45,105,650 gene pairs expression.

The lowest gene pairs count corresponds to the 4(2x2) table type.

Table 2. Number of gene pairs corresponding to each table type in the initial dataset.

<i>Gene1_ClusterCount</i>	<i>Gene2_ClusterCount</i>	<i>TotalCluster_Combination</i>	<i>Gene pairs count</i>
2	2	4	87,471
2	3	6	1,039,097
3	2	6	1,192,266
2	4	8	1,110,036

4	2	8	1,136,671
3	3	9	40,792,519
2	5	10	299,326
5	2	10	320,995
3	4	12	46,729,054
4	3	12	37,320,404
3	5	15	11,170,269
5	3	15	7,368,494
4	4	16	45,105,650
4	5	20	11,153,038
5	4	20	9,234,763
5	5	25	2,603,846

3.3 CLUSTER COMBINATIONS COUNT IN THE INITIAL DATASET

The next step was to count cluster combination categories for each of 16 table types. These results demonstrated prevalent cluster combination categories in each table type based on the quantitative assessment of gene pairs corresponding to each cluster combination category in each table type. The prevalent cluster combination category means the most important number of gene pairs corresponding to this category of cluster combination.

The significant cluster combinations represent different classes of gene dependency relationships between the pairs of genes analysed – which could be driven by a regulatory network (e.g. due to negative feedback) or may indicate cell death due to synthetic lethal relationships. Also, significant cluster combinations indicate at depletion of patient samples in the cells examined for the gene pair evaluated.

The analysis was performed using dplyr library.

There were two versions of the code to produce cluster combinations count. First was “by hand” type of calculation. After preliminary analysis where the information about all existing cluster combination types and table types was found, for each kind of mixture of four columns, Gene1_ClusterCount, Gene2_ClusterCount, TotalCluster_Combinations, Cluster_Combination, the count of gene pairs was executed. This count was made by filtering the four mentioned columns.

Then the automatized version was compiled. First Gene1_ClusterCount, Gene2_ClusterCount, TotalCluster_Combinations, Cluster_Combination columns were grouped and then the summarized number of their mixture was required. The summarized count of these four columns was the number of gene pairs corresponding to each type of these columns’ combination.

Thus, there was a cross verification of results taken from both kinds of analysis.

There were 196 rows in the full result table for cluster combination categories count, which reflects the number of joints between all cluster combination categories with the table types where each of

them presented. It corresponds to the sum of all cluster combinations number in all table types (Σ (4, 8, 8, 9, 10, 10, 12, 12, 15, 15, 16, 20, 20, 25)).

Here the table with the most prevalent cluster combinations for each table type is presented (Table 3). The importance of the cluster combinations in Table 3 was determined by the order of the highest gene pair count for the cluster combination of this table type. The full table and code are placed in the Appendix.

Table 3. The prevalent cluster combination types in each table type in the initial dataset. g1 and g2 with arrows refer to Gene1 and Gene2 levels of expression relatively each other.

<i>Table Cells Number</i>	<i>Table Type</i>	<i>Cluster Combination type</i>	<i>Number of Gene Pairs</i>	<i>Genes Expression Level</i>
4	2x2	2_x_1	43688	g1↑ g2↓
		1_x_2	43724	g1↓ g2↑
6	2x3	2_x_1	487669	g1↑ g2↓
		1_x_2	129404	g1↓ g2↑
		1_x_3	419791	g1↓ g2↑↑
6	3x2	2_x_1	139939	g1↑ g2↓
		3_x_1	479913	g1↑↑g2↓
		1_x_2	568143	g1↓ g2↑
8	2x4	2_x_1	522069	g1↑ g2↓
		1_x_3	358940	g1↓ g2↑↑
		1_x_4	203336	g1↓ g2↑↑↑
8	4x2	3_x_1	362145	g1↑↑ g2↓
		4_x_1	198501	g1↑↑↑g2↓
		1_x_2	552724	g1↓ g2↑
9	3x3	2_x_1	9576825	g1↑ g2↓
		3_x_1	10839289	g1↑↑g2↓
		1_x_2	9427056	g1↓ g2↑
		1_x_3	10876432	g1↓ g2↑↑
10	2x5	2_x_1	132961	g1↑ g2↓
10	5x2	1_x_2	149014	g1↓ g2↑
12	3x4	2_x_1	11683719	g1↑ g2↓
		3_x_1	12007311	g1↑↑ g2↓
		1_x_3	12749708	g1↓g2↑↑

12	4x3	2_x_1	3560338	g1↑ g2↓
		3_x_1	9978144	g1↑↑ g2↓
		4_x_1	4608840	g1↑↑↑g2↓
		1_x_2	9247376	g1↓ g2↑
		1_x_3	9701355	g1↓g2↑↑
15	3x5	2_x_1	2619873	g1↑ g2↓
		3_x_1	2485892	g1↑↑ g2↓
		1_x_3	2162533	g1↓g2↑↑
		1_x_4	2581593	g1↓ g2↑↑↑
15	5x3	3_x_1	1495898	g1↑↑ g2↓
		4_x_1	1583599	g1↑↑↑g2↓
		1_x_2	1761503	g1↓ g2↑
		1_x_3	1735250	g1↓g2↑↑
16	4x4	3_x_1	11930646	g1↑↑ g2↓
		1_x_3	12231568	g1↓g2↑↑
20	4x5	2_x_1	1377793	g1↑ g2↓
		3_x_1	2707848	g1↑↑ g2↓
		1_x_3	2429346	g1↓g2↑↑
		1_x_4	2594170	g1↓ g2↑↑↑
20	5x4	3_x_1	2016170	g1↑↑ g2↓
		4_x_1	1949259	g1↑↑↑g2↓
		1_x_3	2363525	g1↓g2↑↑
		1_x_2	1082245	g1↓ g2↑
25	5x5	3_x_1	531292	g1↑↑ g2↓
		4_x_1	471037	g1↑↑↑g2↓
		1_x_3	555729	g1↓g2↑↑
		1_x_4	542384	g1↓ g2↑↑↑

The results of each cluster combination count showed that in most of the table types there are from two to four important cluster combination categories. Two table types are of exclusion: in both 10-cell table types there is only one the most frequent cluster combination; in 12(4x3) type there are five prevalent cluster combination types.

Cluster combinations which consist of the highest number of gene pairs belong to 9(3x3), 12(3x4) and 16(4x4) table types.

Through the totality of table types all possible classes of cluster combinations were presented by number of gene pairs except “4_x_5” cluster combination of 20(4x5) table type. It means that there was no gene pair which combined clustering created this type of depletion table cell.

Though almost all cluster combinations classes are presented within all amount of table types, not every cluster combination class is presented in each of the table types. The most of cluster combination classes loaded by gene pairs have coordinates with “1” on either first or second position, while “1_x_1” class is not abundant through all table types.

In the general pivot table (in Appendix) with all cluster combination counts and in the table with prevalent cluster combinations (Table 3) the most pronounced quantity have cluster combinations with coordinates “1_x_2”, “1_x_3”, “1_x_4”, “2_x_1”, “3_x_1”, “4_x_1”. These cluster combinations’ coordinates correspond to locations along the edges of the table.

3.4 STANDARDIZING THE TABLE FORM. SWITCHED DATASET.

Next step I unified tables with double-form types to reduce them to one type. There were six table types that were standardized: 6-cell, 8-cell, 10-cell, 12-cell, 15-cell and 20-cell tables. This approach was useful to align the initial dataset for analysis: diminish the number of table types and to optimize further calculations.

The basis of the method used to unify the table types was the matrix transposition strategy. That was possible because the gene pairs expression tables have grid-like data arrangement with rows m and columns n as well as matrices do. The process of matrix transposition involves transforming a matrix by switching its rows and columns.

The process of matrix transposition involves transforming a matrix by switching its rows and columns. The rows of the original matrix become the columns of the transposed matrix. As a result, the matrix is flipped over its diagonal, which runs from the top-left to the bottom-right. The diagonal stays the same, but every element that was in a row moves into a corresponding column. If the original matrix is of size $m \times n$ (i.e., m rows and n columns), the transposed matrix will have size $n \times m$ (i.e., n rows and m columns). The element at position (i, j) in the original matrix is moved to position (j, i) in the transposed matrix. (Strang, 2016)

It should be noted that basically the geometric $(x; y)$ coordinates system was utilized to name the tables. Therefore, tables’ names were “ $n \times m$ ”, so that 2x3 means 2 cells by x-axis and 3 cells by y-axis. However, matrix notation system works in a different way. As a result, the same table in matrix system corresponds to 3x2 ($m=3$ for rows and $n=2$ for columns). This fact was mentioned and explained here for clarity reasons. During the study the geometric coordinates system was employed. (Strang, 2016)

The idea of this mathematical technique was applied to the gene expression tables formed from grids. It was decided to switch all intended tables in the manner that n is smaller than m , for example 3x2 became 2x3 and so on. Instead of the precise numbers there were gene pair counts

corresponding to the “element” in matrix with its cluster combination coordinates analogous to “element position” in matrix. Thus, the following table changes happened: 6(3x2), 8(4x2), 10(5x2), 12(4x3), 15(5x3), 20(5x4) turned to 6(2x3), 8(2x4), 10(2x5), 12(3x4), 15(3x5), 20(4x5) respectively.

Thereby, from 16 table types which presented in the initial dataset rest 10 table types in the new, switched, dataset: 4(2x2), 6(2x3), 8(2x4), 9(3x3), 10(2x5), 12(3x4), 15(3x5), 16(4x4), 20(4x5), 25(5x5).

The process of switching the data of the initial dataset was made using the dplyr library. The main dplyr functions was *mutate* to modify existing columns (Cluster_Combination) and add new transitional columns necessary for switching the data between the columns. The transitional additional columns were created to put in them meanings that should be then switched to another already existing column. The Cluster_Combination column was modified using the determined by *if_else* command conditions described below. The code was compiled to shift meanings between Gene1, Gene2, Gene1_ClusterCount, Gene2_ClusterCount columns and inside Cluster_Combination column according to certain conditions. First, if meaning in the Gene1_ClusterCount column was higher than in the Gene2_ClusterCount, they were switched between these two columns. If this happened, then the Gene1 and Gene2 columns data was also switched due to axis orientation shift. Then, if these two events happened, the switch inside the Cluster_Combination column was executed because of matrix transposition’s rectangular arrangement manner. For example, if there was a cluster combination with “4_x_2” position in the 12(4x3) type of table for gene pair Gene1=ENSG00000003435 and Gene2=ENSG00000002389, after switching the table it would have position coordinates “2_x_4” in the 12(3x4) table type with gene pair Gene1= ENSG00000002389 and Gene2= ENSG00000003435.

Before running the main code, NA rows were eliminated by “na.omit” function. Transitional additional columns were then deleted.

The structure analysis was conducted for the obtained switched dataset in the same manner as for the initial dataset. As far as non-informational rows (NA) were deleted during the switched dataset formation, there were less rows, 216,663,899. As expected, instead of 16, it had 10 table types: 4(2x2), 6(2x3), 8(2x4), 9(3x3), 10(2x5), 12(3x4), 15(3x5), 16(4x4), 20(4x5), 25(5x5). The same 25 categories of cluster combinations were found: “1_x_1”, “1_x_2”, “1_x_3”, “1_x_4”, “1_x_5”, “2_x_1”, “2_x_2”, “2_x_3”, “2_x_4”, “2_x_5”, “3_x_1”, “3_x_2”, “3_x_3”, “3_x_4”, “3_x_5”, “4_x_1”, “4_x_2”, “4_x_3”, “4_x_4”, “4_x_5”, “5_x_1”, “5_x_2”, “5_x_3”, “5_x_4”, “5_x_5”.

As far as tables in switched dataset are unified, the notion of “table form” does not exist anymore. That means that table size and table types are synonyms from this moment.

The pivot table showing gene pairs count for each table type is Table 4.

Table 4. Number of gene pairs corresponding to each table type in the switched dataset.

<i>Gene1_Cluster Count</i>	<i>Gene2_Cluster Count</i>	<i>TotalCluster_Combination</i>	<i>Gene pairs count</i>
2	2	4	87,471
2	3	6	2,231,363
2	4	8	2,246,707
3	3	9	40,792,519
2	5	10	620,321
3	4	12	84,049,458
3	5	15	18,538,763
4	4	16	45,105,650
4	5	20	20,387,801
5	5	25	2,603,846

As observed, all table types that were standardized began to consist of a greater number of genes exactly equal to the number contained in the eliminated table type. This fact can additionally indicate that the standardization process passed correctly. It is important to note that there were no changes in the columns' names and numbers of the switched table.

The highest count of gene pairs matches the 12-cell table type (84,049,458), then follows 16-cell table type (45,105,650) and 9-cell table type (40,792,519).

3.5 CLUSTER COUNT IN THE SWITCHED DATASET

After standardization modification of the initial dataset, the new count of cluster combination categories in each of 10 table types was executed for switched dataset in the same manner as for the initial dataset. Quantification results showed augmentation of the gene pairs number for each cluster combination category in all table types except 4-cell, 9-cell, 16-cell and 25-cell kinds. That matches to the similar effect in table types, when in the unified kinds of tables more gene pairs appeared.

The code used for that step was the same as was used for the cluster combinations count in the initial dataset as far as the switched dataset had the same columns structure as the initial one. Only the automatized version was applied.

The most pronounced categories to which the prevalent number of gene pairs related were found in the 9-cell, 12-cell and 16-cell type tables as well as in the initial dataset.

The prevalent cluster combinations now occupied 125 rows because of the reduction of cluster combination categories and table type joints after elimination of some table types. The sum of all cluster combinations in all table types is now presented as $\Sigma (4, 8, 9, 10, 12, 15, 16, 20, 25)$.

Here the table with the dominant cluster combination categories for each table type is presented (Table 5). The code and full table are in the Appendix.

Table 5. The prevalent cluster combination types in each table type in the switched dataset. g1 and g2 with arrows refer to Gene1 and Gene2 levels of expression relatively each other.

<i>Table Cells Number</i>	<i>Table Type</i>	<i>Cluster Combination type</i>	<i>Number of Gene Pairs</i>	<i>Genes Expression Level</i>
4	2x2	2_x_1	43688	g1↑ g2↓
		1_x_2	43724	g1↓ g2↑
6	2x3	2_x_1	1055812	g1↑ g2↓
8	2x4	2_x_1	1074793	g1↑ g2↓
9	3x3	2_x_1	9576825	g1↑ g2↓
		3_x_1	10839289	g1↑↑g2↓
		1_x_2	9427056	g1↓ g2↑
		1_x_3	10876432	g1↓ g2↑↑
10	2x5	2_x_1	281975	g1↑ g2↓
		1_x_4	166208	g1↓g2↑↑↑
12	3x4	2_x_1	20931095	g1↑ g2↓
		3_x_1	21708666	g1↑↑ g2↓
		1_x_3	22727852	g1↓g2↑↑
		1_x_4	10830385	g1↓ g2↑↑↑
15	3x5	2_x_1	4381376	g1↑ g2↓
		3_x_1	4221142	g1↑↑ g2↓
		1_x_3	3658431	g1↓g2↑↑
		1_x_4	4165192	g1↓ g2↑↑↑
16	4x4	3_x_1	11930646	g1↑↑ g2↓
		1_x_3	12231568	g1↓g2↑↑
20	4x5	2_x_1	2460038	g1↑ g2↓
		3_x_1	5071373	g1↑↑ g2↓
		4_x_1	1676220	g1↑↑↑g2↓
		1_x_3	4445516	g1↓g2↑↑
		1_x_4	4543429	g1↓ g2↑↑↑
25	5x5	3_x_1	531292	g1↑↑ g2↓
		4_x_1	471037	g1↑↑↑g2↓

		1_x_3	555729	g1↓g2↑↑
		1_x_4	542384	g1↓g2↑↑↑

The prevalence of cluster combinations in Table 5 was based on the order of the highest gene pair count same as for cluster combinations count in Table 3.

Cluster combination count in the switched table has more classes of important cluster combinations in most table types. Tables have from one to five of prevalent cluster combinations categories. 20-cell table has the highest number of important cluster combination categories equal to five, while 6-cell and 8-cell tables have one prevalent cluster combination each.

For switched dataset the same manner of cluster combination classes distribution repeats. The prevalent classes of cluster combinations most often have coordinates with “1” on first or second coordinate's positions with no prevalence for “1_x_1” class of cluster combinations. Therefore, the largest number of gene pairs within different table types refer to cluster combinations such as “1_x_2”, “1_x_3”, “1_x_4”, “2_x_1”, “3_x_1”, “4_x_1” in both general pivot table in Appendix and in the Table 5 with prevalent cluster combinations.

After standardization there are no cluster combinations with gene pairs count equal to zero. Another point that happened after dataset standardization is that all table types started to contain all classes of cluster combinations. Thus, they assembled in one table size after unifying table forms.

3.6 TOTAL ASSEMBLAGE OF CLUSTER COMBINATIONS: PATTERN DATASET. UNIQUE PATTERNS.

One of the key results of the present dataset analysis was to find combinations of all the significant table cells for each gene pairs to determine the composite GDR pattern for the gene pair. The resulting pattern dataset table contained 53,996,489 rows and consisted of the following columns: Gene1, Gene2, Gene1_ClusterCount, Gene2_ClusterCount, TotalCluster_Combinations, Result_Comb. Gene1, Gene2, Gene1_ClusterCount, Gene2_ClusterCount and TotalCluster_Combinations columns embodied the same information as in the initial dataset. Result_Comb column includes the combinations of the cluster combinations, that are called patterns in the text below. The number of rows in the pattern dataset diminished because repeated gene pairs from the initial dataset were reduced. This happened after combining different cluster combinations related to the same gene pair into the pattern. One pattern can correspond to different gene pairs, but for each gene pair there is only one pattern. For that reason, it is necessary to find unique patterns along with the count of gene pairs associated with these patterns. Additionally, a conclusion can be made that the pattern dataset represents the list of the unique gene pairs from the initial dataset.

To create pattern dataset both dplyr and tidyverse libraries were used and the next code was compiled. First, the columns of interest were selected to save them in a new dataset and to work with them: Gene1, Gene2, Gene1_ClusterCount, Gene2_ClusterCount, TotalCluster_Combinations, Cluster_Combination and the duplicates were eliminated from them. Then these columns were grouped to be summarized by Cluster_Combination meanings which were accumulated in rows with the help of tidyverse library functions *str_c* and *collapse*. The new Result_Comb column appeared. The rows were arranged by Cluster_Combination and Result_Comb columns' values. Additionally, the patterns in the Result_Comb column were ordered by the length of the row defined by the number of cluster combinations contained by each pattern. Thus, in pattern dataset in Result_Comb column first appear all patterns with single cluster combinations starting from "1". After single cluster combination pattern, numerous cluster combinations patterns follow. In patterns structured from two and more cluster combinations first come cluster combinations which start from "1", then from "2" and so on. It is important to note that one cluster combination can also represent a pattern.

The whole pattern dataset was divided into 10 sub-datasets for each table type. This was made both for conceptual and technical reasons. Conceptually, the different table types were created by gene pairs that can create a mixture of a certain number of clusters according to their expression behavior in samples. However, when individual genes combine into cluster combinations (table cells) and total cluster combinations (tables), they create new clustering landscape corresponding to their co-expression in the samples. That means that, for example, the gene which expression produces two clusters generates different composite of clustering each time when it cooperates with genes that generate two or three or four clusters. Thus, the same gene would behave in a distinct manner and create a new expression profile depending on his partner gene. Consequently, a pattern created from a set of cluster combinations in the 4-cell table type is different from a pattern that was created from the indexed cluster combinations in the 6-cell table type. For example, "2_x_1" cluster combination in 4-cell table type is not the same as those in the 6-cell table type, although they have same table cell coordinates.

To create the pattern sub-datasets filtering of the TotalCluster_Combinations column by table type was employed. Then arranging of the rows in ascending order of first values and number of elements in the group was executed by *arrange* and *nchar* functions. This was necessary to perform because these datasets were new, and the order of the big pattern dataset was not kept in them without applying these functions. From each of the pattern sub-datasets was created one-column files with a list of unique patterns which then was counted.

Each of 10 pattern sub-datasets was analyzed on the total amount of gene pairs implicated in the production of patterns for that table type. In other words, gene pairs that are unique for certain table type patterns. Then, the unique patterns lists were created for each of the table type together with the count of each unique pattern. Full lists of the patterns are in the Appendix. For further patterns analysis only three first patterns were taken among those that were created by the highest number of gene pairs from all table types, except the 25-cell table type where 10 leader

patterns were chosen. They were held under graphical analysis the result of which is represented in Figure 3. (Table 6; Figure 3)

Table 6. Pattern dataset analysis: count of pattern types and gene pairs creating patterns in each table type.

<i>Table types (by number of cells)</i>	<i>Number of gene pairs in table type</i>	<i>Number of unique pattern types</i>
4	37,178	6
6	863,247	18
8	899,935	26
9	9,726,522	80
10	246,161	75
12	20,432,259	223
15	4,608,393	493
16	11,181,031	568
20	5,295,774	1553
25	705,989	2501

The highest numbers of gene pairs correspond to 12-cell, 16-cell and 9-cell table patterns. 15-cell and 20-cell table patterns are also constructed by a high number of gene pairs. The lowest number of gene pairs are engaged in 4-cell table patterns. However, the general trend indicates an increase in the number of pattern types as the number of table cells grows.

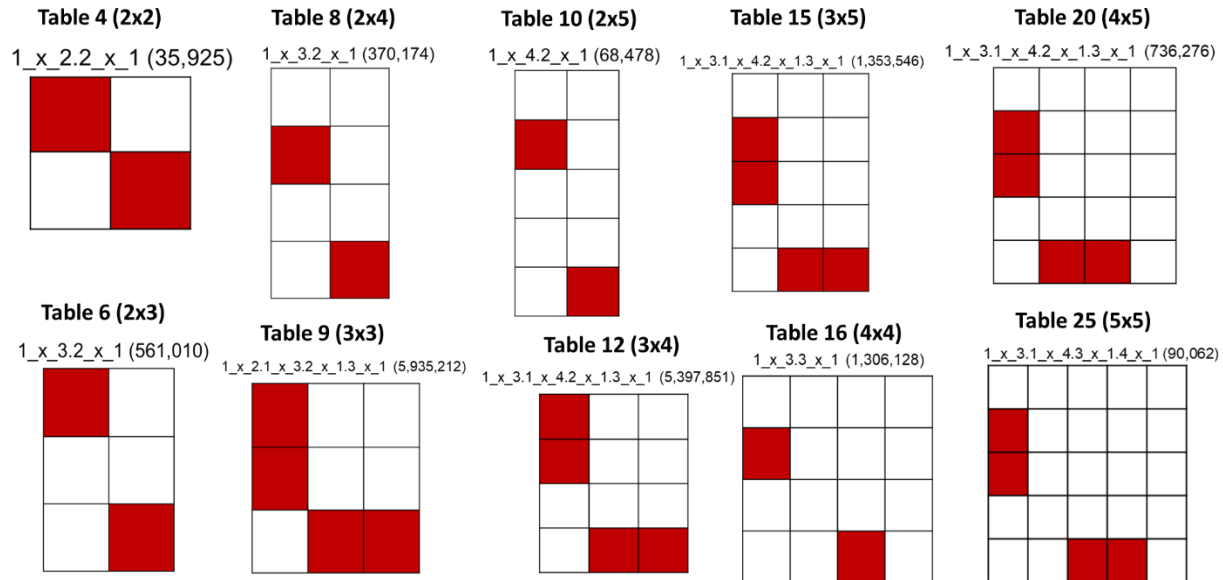


Figure 3. Prevalent patterns visualization. Patterns from each table type which were created by the highest number of gene pairs (prevalent patterns of leader patterns). The table type of the pattern, the mixture of prevalent cluster combinations of the pattern and the number of gene pairs that form this pattern are indicated above each pattern image.

The composition of leader patterns (37 patterns visualized) is mostly represented by the cluster combinations categories such as “1_x_2”, “1_x_3”, “1_x_4” and “2_x_1”, “3_x_1”, “1_x_4”. That mirrors the same phenomenon appeared on the cluster combinations level, where cluster combinations with the same type of coordinates had the predominant count.

It can be noticed that among the prevalent patterns among the leader patterns (on Figure 2) there is a part that has two cluster combinations, and another part has four cluster combinations. Patterns that contain two cluster combinations belong to 4, 6, 8, 10, 16 table types. Four cluster combinations were found in 9, 12, 15, 20, 25 table types. Thus, the prevalent patterns of the leader pattern classes with four cluster combinations are formed by gene pairs where one of genes creates either three or five clusters.

However, this is not applicable to all the 37 patterns visualized. For example, 6-cell, 8-cell and 25-cell patterns are highly variable in terms of cluster combinations number that they include, while 9-cell, 12-cell and 20-cell patterns are rather stable for this parameter.

It is noticeable that all cluster combinations are located on the edges of the table. This is true for all table types of the leader pattern classes. Appendix.

3.7 GENE PAIRS LISTS AND SINGLE GENE LISTS

From each pattern sub-dataset containing unique gene pairs for the table type were created lists with gene pairs which are unique for each pattern of the table type. Then, from the gene pairs lists single genes were extracted. This analysis was applied for the same patterns which were previously visualized.

To create gene pairs lists pattern sub-dataset of each table type was filtered by the pattern of interest with checking the count of gene pairs obtained. Then, this unique gene pairs list was treated with *tidyr* library *pivot_longer* function was applied to unite genes from two columns into one. Then, using *dplyr* library functions *select*, *distinct* and *arrange* the transitional name column was deleted, duplicated genes were eliminated, and finally, single genes were arranged in an ascending manner. The number of single genes was evaluated as well.

The number of gene pairs corresponding to every visualized pattern was counted as well as the number of single genes. (Table 7)

Table 7. Unique gene pairs and unique single genes count for the prevalent patterns of each table type.

<i>Table type</i>	<i>Gene pairs count for table type</i>	<i>Pattern</i>	<i>Gene pairs count for pattern</i>	<i>Single genes count</i>
4	37,178	1_x_2.2_x_1	35,925	2,237
		2_x_1	604	486
		1_x_2	601	461

6	863,247	1_x_3.2_x_1	561,010	9,308
		1_x_2.1_x_3.2_x_1	170,230	6,956
		2_x_1	78,768	7,440
8	899,935	1_x_3.2_x_1	370,174	10,891
		1_x_3.1_x_4.2_x_1	201,125	8,307
		2_x_1	166,743	10,410
9	9,726,522	1_x_2.1_x_3.2_x_1.3_x_1	5,935,212	6,577
		1_x_3.2_x_1.3_x_1	1,058,576	6,704
		1_x_2.1_x_3.3_x_1	953,487	6,729
10	246,161	1_x_4.2_x_1	68,478	5,545
		2_x_1	54,806	5,455
		1_x_3.1_x_4.2_x_1	31,713	3,553
12	20,432,259	1_x_3.1_x_4.2_x_1.3_x_1	5,397,851	13,826
		1_x_3.2_x_1.3_x_1	4,517,289	15,830
		1_x_2.1_x_3.2_x_1.3_x_1	3,508,021	13,044
15	4,608,393	1_x_3.1_x_4.2_x_1.3_x_1	1,353,546	9,852
		1_x_3.2_x_1.3_x_1	436,924	9,571
		1_x_4.2_x_1.3_x_1	307,104	9,834
16	11,181,031	1_x_3.3_x_1	1,306,128	9,435
		1_x_3.1_x_4.2_x_1.3_x_1	1,196,512	8,720
		1_x_2.1_x_3.3_x_1.4_x_1	961,519	8,677
20	5,295,774	1_x_3.1_x_4.2_x_1.3_x_1	736,276	11,061
		1_x_3.1_x_4.3_x_1	594,581	12,585
		1_x_3.1_x_4.3_x_1.4_x_1	346,043	9,246

25	705,989	1_x_3.1_x_4.3_x_1.4_x_1	90,062	3,448
		1_x_3.1_x_4.3_x_1	49,810	3,637
		1_x_3.3_x_1.4_x_1	38,892	3,564
		1_x_4	30,358	3,936
		1_x_3.1_x_4	24,192	3,486
		1_x_3.1_x_4.4_x_1	21,058	3,263
		1_x_4.3_x_1	18,708	3,587
		1_x_3.1_x_4.2_x_1.3_x_1	18,288	2,522
		1_x_3.4_x_1	16,381	3,610
		1_x_4.3_x_1.4_x_1	16,344	3,182

Comparison of the “Gene pairs count for pattern” and “Single genes count” columns in each table type showed that even when there was rather noticeable difference in counts between patterns’ gene pairs there is still homogenous count level for single genes inside the table type. Additionally, if there is a high number of gene pairs it does not mean the high number of single genes which form these gene pairs.

The highest number of gene pairs correspond to pattern classes of the 9-cell, 12-cell, 15-cell, 16-cell table types, among which 12-cell table contains the greatest quantity of gene pairs. The highest number of single genes corresponds to pattern classes of the 8-cell, 12-cell and 20-cell table types, among which 12-cell the greatest single genes count.

3.8 DIFFERENCE BETWEEN SINGLE GENE LISTS

Due to the vast quantity of found single genes, it was decided to search for lists of common and different single genes to make these lists more selective and, thus, to identify implicated biological functions more precisely.

This was made for part of the tables: from 4-cell to 20-cell sized tables.

To find common single genes tidyverse’s *purrr* library function *reduce* was used together with dplyr’s function *inner_join*. These functions can be applied to compare several lists at the same time inside each table type. To find different single genes *setdiff* function of the base R library was applied. This time, single gene lists should be compared by one pattern class with other pattern classes in each table type.

Table 8. Common and different genes between single genes lists of the same table type.

<i>Table type</i>	<i>Pattern</i>	<i>Number of common genes</i>	<i>Number of different genes</i>
4	1_x_2.2_x_1	204	1580
	2_x_1		25
	1_x_2		29
6	1_x_3.2_x_1	5831	1339
	1_x_2.1_x_3.2_x_1		152
	2_x_1		306
8	1_x_3.2_x_1	7018	727
	1_x_3.1_x_4.2_x_1		347
	2_x_1		624
9	1_x_2.1_x_3.2_x_1.3_x_1	6306	92
	1_x_3.2_x_1.3_x_1		128
	1_x_2.1_x_3.3_x_1		96
10	1_x_4.2_x_1	2909	609
	2_x_1		779
	1_x_3.1_x_4.2_x_1		190
12	1_x_3.1_x_4.2_x_1.3_x_1	11025	335
	1_x_3.2_x_1.3_x_1		543
	1_x_2.1_x_3.2_x_1.3_x_1		161
15	1_x_3.1_x_4.2_x_1.3_x_1	8715	240
	1_x_3.2_x_1.3_x_1		317
	1_x_4.2_x_1.3_x_1		403
16	1_x_3.3_x_1	8288	484
	1_x_3.1_x_4.2_x_1.3_x_1		38
	1_x_2.1_x_3.3_x_1.4_x_1		40

20	1_x_3.1_x_4.2_x_1.3_x_1	7746	322
	1_x_3.1_x_4.3_x_1		658
	1_x_3.1_x_4.3_x_1.4_x_1		214

After the comparative analysis between single gene lists of patterns belonging to different table types, the number of genes reduced, especially in the lists of difference. Most of the lists of differences consist of hundreds of genes, which are suitable for further functional biological analysis and can bring more essential information. Thus, this last step made during the present study prepared gene lists that are ready for further dissection.

IV. DISCUSSION

One of the ways to study gene dependency relationship (GDR) is to analyze the distribution of sample clusters based on the expression ratio of a gene pair (or more than two genes). In the present research, the relationship between two genes was analyzed with two aims. First, was to find repeating clustering structure called pattern constructed from certain categories of cluster combinations. Second, to create a list of gene pairs and single genes which participate in creating each cluster combinations' configuration type. These lists were then planned to analyze in gene functional annotation tools such as DAVID, KEGG and Reactome databases. (Sherman et al., 2022; Kanehisa, 2023; Rothfels et al., 2023)

During the dataset analysis several levels of clustering organization in gene pairs of multiple myeloma samples were studied: cluster combinations level, clustering table level, pattern level. There are some trends that can be deduced from the prevalent cluster combinations, table and pattern types.

Additionally, some interesting tendencies of gene pairs counts distribution on different levels of expression clustering organization such as cluster combinations, tables and patterns were found. There are also some regularities in the count distribution between gene pairs and through single genes for different types of patterns. This part of the results also includes common and different genes between different patterns in one table type.

The prevalent patterns' structures were visualized and the tendency of cluster combinations assembly in patterns was detected. These results shed some light on the GDR type inherent to the gene pairs from the analyzed multiple myeloma samples.

After the clustering analysis including the discovery of pattern general repeating motifs in all table types lists of gene pairs and genes were extracted which are expected to have biological meaning and thus, can be considered as potential multiple myeloma drug targets.

4.1 RESULTS INTERPRETATION

4.1.1. Preliminary analysis of the initial dataset

The Myeloma_XPR_GDR dataset presented a result of the analysis of the transcriptional expression data of the 928 patients diagnosed with multiple myeloma from the IA12 release of the CoMMpass study on the MMRF site (<https://themmr.org/>; Shaw, 2021; Settino, 2021). This is important to mention because the CoMMpass precursor dataset passed through complex MultiSEp approach (Wappett et al., 2021) and further statistical analysis so that the gene expression data was transformed. This fact had consequences in terms of the analytic approaches that could be used for the data treatment. First, the Myeloma_XPR_GDR dataset presented for analysis in the actual study contained only gene pairs with statistically approved biological effect through multiple myeloma samples. Second, the biological effect was a depletion of samples resulting from the gene interaction inside the gene pairs. This could be seen from the Expected_Count and Actual_Count

columns which were not directly used during the present analysis but demonstrated less counts in actual counts than in the expected counts. Thus, the depletion phenomenon found for all gene pairs in the Myeloma_XPR_GDR dataset was statistically significant for every gene pair co-expression clustering and for all analyzed clustering organization levels such as cluster combinations, tables and patterns.

After preliminary analysis of the Myeloma_XPR_GDR dataset, it became clear that all possible table types and sizes are presented considering the rules of the table's formation (more than two clusters from each gene, from two to five clusters from each gene in the gene pair, tables cannot be created by prime numbers). For all table types where two table forms were possible (6, 8, 10, 12, 15, 20), both forms existed, which means that there was a certain number of gene pairs which co-expression created these tables.

These facts seem to be related to the statistical significance of gene pairs that cause depletion in the clustering table structures. This also suggests that gene pairs from the Myeloma_XPR_GDR dataset are somehow involved in multiple myeloma biology.

The fact that there were two-form tables for some table sizes can occur because one and the same gene can be either Gene1 or Gene2 depending on the gene pair. Even if the gene pairs are different each single gene from gene pair can participate in a gene pair with another gene. Thus, it can be in pair either with gene which create more clusters, or with those that create less clusters.

The summarizing result of gene pairs count for all table types on the stage of the preliminary analysis showed that the prevalent number of gene pairs belong to the five table types which are 12-cell, 16-cell, 9-cell, 20-cell, 15-cell types (listed from most to least numerous). Further analysis suggested some explanations of this outcome. Additionally, in each prevalent two-formed table type (12-cell, 20-cell, 15-cell) for both table forms of the same table size high gene pairs counts were inherent.

It is also important to mention about the order of the gene pairs quantity which was mostly more than one million gene pairs for a table type and for some tables were formed by expression clustering of tens of millions of gene pairs. Further, this evidence would affect the strategy of gene pairs and single genes lists compilation and their subsequent analysis options. However, gene pairs count was not evenly distributed among different table types and there was no relation between the size of the table and the number of gene pairs that it was formed by. Thus, the cluster combinations count was performed for deeper analysis of combined clustering elements formation which reflects regularities of gene expression interactions and gene dependency relationships.

4.1.2. Cluster combinations count

Cluster combination count meant the number of gene pairs that corresponded to the certain cluster combination of a certain table type. In other words, it showed gene pairs partition between different cluster combinations inside each group of table types.

In this section both cluster combination counts are discussed, before and after dataset standardization. As was mentioned in the results chapter, after standardization in the tables with unequal number of table cells by x-axis and y-axis all cluster combination classes from eliminated table types assembled in the table types that remained. The same happened with the number of gene pairs which gathered in remaining table types.

Importantly, before and after standardization of the dataset the same tendency was kept. In gene pairs of prevalent cluster combinations one of genes has “1” position and another gene has “2”, “3” or “4” position. The combination of clusters of two genes correspond to their co-expression in the studied tables. Thus, the coordinates reflect the level of expression for each of two genes. Returning to the tendency that was noticed in the cluster combination classes distribution, it shows that within all the most frequent cluster combinations two genes always have opposite levels of expression. Thus, in the scale of one cluster combination, among the most frequent of its categories, one of two genes always has lower expression level than its pair. This was first found in the Myeloma_XPR_GDR dataset. There is a similar trend for the switched dataset. The prevalent cluster combinations always have the inverse gene expression levels for each of two genes in a gene pair, when one of them has lower expression level than another one.

To demonstrate this event, there is a column in the general pivot tables in Appendix where the levels of expression for each gene in gene pair is indicated relative to each other.

Another information that was provided by the cluster combination count was that on the cluster combinations level the same distribution of gene pairs number was found. The prevalent cluster combinations count was found in the 12-cell, 16-cell and 9-cell table types (order is indicated from greatest to smallest) which explains the prevalent table types found during preliminary Myeloma_XPR_GDR dataset analysis. After dataset standardization this prevalence remained.

These results can indicate that most genes from gene pairs in the studied dataset produce mainly 3 and 4 clusters of multiple myeloma samples. This can be deduced from the fact that 9, 12 and 16 are produced from multiplication of 3 by 3, 3 by 4 (or 4 by 3) and 4 by 4. Thus, the combining of these genes in gene pairs generates gene co-expression tables that have 9-cell, 12-cell and 16-cell table size. Additionally, the table types which are partly formed by the genes creating 3 or 4 clusters, like 15-cell (3x5) and 20-cell (4x5) tables, are also highly represented.

However, the biological reasons of this occurrence are not clear as well as its degree of functional impact. In future studies it could be interesting to compare biological role and functional interactions of the gene pairs and genes which form these expression tables between each other and with the rest of the table types. This can help to detect the core genes and gene dependencies that are crucial to multiple myeloma pathological functioning.

4.1.3 Patterns

The leader patterns that were found and visualized confirm and conclude the trends that appeared during the preliminary analysis of table types and the cluster combination counts. The highest count of gene pairs corresponds to the patterns from 12-cell, 9-cell and 16-cell table types which includes “1_x_2”, “1_x_3”, “1_x_4”, “2_x_1”, “3_x_1” and “4_x_1” cluster combinations categories. These trends have two consequences.

First, distribution of gene pairs’ number between pattern classes inside one table type. The number of cells that each table type consists, the number of gene pairs involved in a table type and the number of unique patterns for each table type generate a special distribution of gene pair counts between patterns inside one table type. This distribution is predetermined basically by the number of gene pairs forming the most often cluster combinations’ categories and their final arrangement in patterns and consequently revealed during the pattern analysis because of that fact. Incidentally it leads to the effect when table types with big number of table cells, lots of possible variants of cluster combinations, and thus, a big number of unique pattern types, but relatively low number of gene pairs included have a smear of pattern classes consisting of only one gene pair. This event is particularly evident in 20-cell and 25-cell table types. Therefore, it is not clear how to evaluate the impact of these poorly pronounced patterns and where is the threshold of their functional importance.

Second, the cluster combinations distribution behavior inside the leader patterns. The distinctive character of cluster combinations disposition in the leader patterns was especially obvious after visualization analysis. It showed the peripheral location of the cluster combinations in the leader patterns. This is logically connected with the cluster combination count results. The leader patterns are constructed from the most abundant cluster combinations which had the coordinates that refer to this border location.

From a biological point of view, patterns present a certain type of interaction between genes repeating within a several number of gene pairs. This relationship between genes is implemented on the transcription level and reflected through their expression manner, which includes the expression level ratio between the genes. The biological event happens as a result of gene relations and following gene expression. The biological functional outcome of interest in the present study is sample depletion.

From methodological point of view, pattern is created by cluster combinations formed by numerous gene pairs with the same type of communication between them resulting in depletion effect. In other words, depletion happens due to genes’ co-expression which can be observed in the pattern table as a merge of two genes’ clustering, or cluster combination. The position, or coordinates, of this clustering merge of two genes indicates the level of expression for each of two genes.

Thus, the location of cluster combinations in the pattern shows both genes' simultaneous expression state or expression level ratio while depletion event happens because of co-expression of these two genes from a gene pair.

An evident motif of visualized patterns formed by edge-positioned cluster combinations demonstrates that in the involved gene pairs two genes have opposite levels of expression. The overlap of two genes' clustering appears in the table points where one gene is significantly lower expressed than another gene in the pair. Thus, the co-expression pattern of these two genes exhibits an inverse relationship.

To summarize, the visualized patterns show that depletion in multiple myeloma samples happens when two genes with opposite expression levels are co-expressed following certain biological regulations. This biological control system suggests a potential negative regulatory interaction between the two genes, indicative of a mutual inhibitory mechanism. Such a regulation can appear in the case of negative feedback loop network motifs when activation of one component leads to the repression of another helping system stabilize or create alternating states. The opposing expression levels may point to a functional antagonism between these genes, possibly reflecting their involvement in distinct regulatory pathways. Further validation such as pathway analysis would help clarify the exact nature of this inverse relationship. (Alon, 2007; Li, 2014; Lee et al., 2014; Mossmann, 2018; Yulan et al., 2019; Zou et al., 2020; Tang et al., 2024)

4.1.4 Gene lists

There was only quantitative analysis of gene pairs' lists and single genes' lists. That is why it is hard to give pure biological interpretation of this part of analysis. However, the numerical information found can give some indirect inferences about functional interactions of the found genes.

The difference between gene pair count and single gene count can indicate the common level of interaction potential of genes with each other through gene pairs within one pattern. For example, there were patterns created from gene pairs that consisted of a smaller number of single genes than another gene pairs while generating more gene pairs that finally formed this pattern. This suggests that, despite the lower number of single genes, these genes are much more functionally interconnected with each other than those that form a smaller number of gene pairs.

The level of interaction between genes could also be analyzed by matching gene pairs count for pattern, single genes count for one table type and the number of common genes between different patterns of one table type. In this case, common genes could be indicators of the interconnection strength between the genes of different patterns in one table type.

The calculations of common and different single genes between patterns in the same table type can indicate the contribution of each cluster combination in common number of single genes involved

in one pattern. Thus, this appears to show possible important gene pairs which are formed by these genes and their co-expression dependency.

However, all these hypotheses need to be confirmed by further analysis, including the examination of if and how found trends are related to the table cell number of the patterns.

4.2 CONSEQUENCES AND IMPLICATIONS

Network motifs of transcriptional regulation, gene dependencies relationships and how they form tumor pathogenesis have been studied during several decades mainly with the aim to find the most effective targets for anti-cancer drugs. (Darnell, 2002; Ravasi et al., 2010; Bushweller, 2019)

Gene dependencies relationships are closely interconnected with transcriptional network regulations motifs. While gene dependencies create some expression print of clustering, transcriptional regulation motifs can indicate type of gene dependency relationships which affects the involved biological processes. Transcription is conducted by network regulatory motifs which are recurring regulation patterns. These regulatory patterns are repeated circuits of gene interaction during the transcription creating functional network that determine which biological effects will be finally achieved: cell death, cell alive, different cell functioning modes like depletion. (Alon, 2007; Gao, 2009)

Along with the search for cancer driven mutations (ICGC/TCGA Pan-Cancer Analysis of Whole Genomes Consortium, 2020), there were always attempts to find systemic interactions particularly in regulation of gene transcription (Awan et al., 2007; Carson, 2015; Attwood, 2021; Dehghan, 2022; Kumar, 2023; Leblay et al., 2024). By affecting these interactions more powerful cancer treatment could be developed. This idea formed the basis of molecular vulnerabilities search. Mutations in cancer cells often create distinct molecular vulnerabilities that are absent in normal cells. These vulnerabilities, known as "Achilles Heels," present unique opportunities for targeted therapeutic interventions. By identifying specific dependencies in cancer cells that arise from their mutations, more precise drug targets aimed at exploiting these weaknesses. Several large-scale initiatives, such as the Cancer Cell Line Encyclopedia (CCLE) and the Cancer Dependency Map (DepMap), have been instrumental in cataloging and analyzing cancer-specific genetic dependencies. These and similar projects aggregate various multiomics data and offer comprehensive repositories of genetic vulnerabilities, specific mutations and sensitivities to therapeutic agents. (Tsherniak et al., 2017; Barretina et al., 2012; Wappett, 2021).

The gene dependencies form several types of relationships understanding for which is essential for even deeper understanding of cancer molecular vulnerabilities (Pacini et al., 2024). Among the GDR synthetic lethality is especially promising as an anti-cancer drug target. PARP inhibitors used for cancer treatment are a classic example of synthetic lethality relationships exploitation (Lord, 2017; Li, 2022).

Several studies were dedicated to investigating gene dependencies and transcriptional regulatory motifs in multiple myeloma (Ziccheddu et al., 2021; Went, 2022; Walker et al., 2018), including synthetic lethality mechanism (Xu et al., 2019; Botrugno et al., 2020; Teoh et al., 2022). Still this hematological malignancy remains understudied.

In terms of targetable biological systems, the ubiquitin-proteasome system is generally promising for cancer treatment and plays a central role in the development of multiple myeloma treatments. Proteasome inhibitors form the cornerstone of therapeutic approaches for multiple myeloma. (Ito, 2020; Zhou, 2023) However, there are quite a few scientific papers where gene dependency such as synthetic lethality and ubiquitin proteasome system were studied together (Neri et al., 2011).

The actual study planned to find GDR type, presumably synthetic lethality, within NRM patterns in multiple myeloma gene pairs transcriptional dataset with possible involvement of ubiquitin proteasome system genes. Not all projected stages of the study were executed. The obtained information mainly resulted in transcriptional patterns and in lists of genes with some quantitative analysis. The found regulatory motif of mutual inhibition between two genes where possible recurring regulation pattern is negative feedback loop was already described in the literature. The feedback loop regulation motif is important for multiple myeloma progression (Sun, 2014; Xiao, 2019; Vega, 2022). Nevertheless, it does not obviously underly the synthetic lethality GDR. It mostly shows the general NRM that characterizes relationships of the studied genes on expression level in multiple myeloma samples. Thus, additional analysis is necessary to classify these regulatory interactions more precisely.

Discovering the functional interactions and biological roles of the genes in both gene pairs and single gene lists could provide deeper insights into GDR mechanisms.

The highly expressed genes in multiple myeloma mentioned in the literature are MCL1, CXCR4, TNFRSF17, SDC1, SLAMF7, PTP4A3, XBP1, IFI6, IFITM1, PTP4A3, SIK1, ALDOA, ATP5MF, ATP5ME, and PSMB4 (Børset et al., 2022). The most often altered genes in multiple myeloma are translocations in MAF, MAFB, CCND3, CCND1, FGR3/MMSET, MYC; chromosome loss TP53, CDKN2C or FAM46C, CD27; chromosome gains MCL1, CKS1B, ANP32E or BCL9. (Zhan, 2006; Castaneda, 2019). In the present research genes from the analyzed dataset were not annotated and there was only their quantitative characterization.

Among the recent literature there are sources dedicated to myeloma related genes, their functional impact with further drug target prospect. In these papers DepMap data and CoMMpass datasets were employed. They showed FABP, TCF3, FLVCR1, IKBKB, PIM2, CHUK and RELB genes are strongly involved in MM progression and maintaining tumor viability. Thus, these genes are considered as promising targets for MM therapy. (Went, 2022; Farrell, 2023)

TP53 presents the classical example of tumor suppressor gene and MYC is a classical oncogene, which are both implicated in the multiple myeloma. Their interactions could be one of a possible example of mutual inhibition relationships implemented by negative feedback loop regulation mechanism (Zhan, 2006). This common example was given as a possible explanation of the results

issued from the actual analysis. It is evident that identifying more specific genes involved in multiple myeloma would lead to a better understanding of the mutual inhibition and negative feedback loop regulatory mechanisms in gene dependencies of the disease.

V. CONCLUSIONS

Initially, the aim of the present study was to find genes and gene pairs involved in the prevalent gene dependency relationships (GDR) type and identify the category of this GRD in present collection of multiple myeloma samples. It was partially implied that the GDR type could refer to the synthetic lethality. Quantitative analysis was applied to the different structures of sample clustering resulting from transcriptional expression. This was made to uncover the principal network regulatory motif that can indicate the kind of gene dependency relationships which affects the involved biological processes. The biological process of interest was ubiquitin proteasome system, a cellular mechanism which is a prospective drug target for cancer treatment.

5.1 KEY FINDINGS

- 1) Gene pair co-expression in the studied multiple myeloma dataset predominantly resulted in the formation of 3 to 4 sample clusters.
- 2) A clear trend of depletion patterns emerges in multiple myeloma samples, with most gene pairs forming cluster combinations where one gene has lower expression than the other.
- 3) The found type of network regulatory motif is the most closely matches feedback loop based on mutual inhibition. This transcription regulatory motif can potentially be underly synthetic lethality-type dependencies. However, the precise mechanism of its implementation requires further investigation.
- 4) Gene pairs lists and single gene lists of the prevalent patterns were created for further gene ontology enrichment and functional annotation clustering analyses.
- 5) The number of common genes is much greater than the number of different genes between prevalent patterns of the same expression table type.

5.2 FINAL THOUGHTS

During the present study some challenges happened mainly on the stage of creating and analyzing gene pair lists and single genes' lists.

First, the number of genes in the lists were excessively high and above optimal levels to put them in the computational tool for gene enrichment and functional annotation. Second, due to technical constraints, gene lists were not subjected to functional analysis. They passed only through quantitative analysis and no information about functional biological significance and interactions was obtained. Thus, only common conclusions based on the numerical data could be reached.

Additionally, for the same technical reasons gene pairs' lists were not studied properly, and only quantitative results were provided.

5.3 RECOMMENDATIONS

To develop the results of the present study further computational analysis should be implemented: SynLeGG, NetNC and Cytoscape to approve actual results and determine the validity of a synthetic

lethality hypothesis; DAVID to find biological significance of genes from the single gene lists; Cytoscape, Reactome and KEGG to investigate functional interactions inside gene pairs. (Sherman et al., 2022; Kanehisa, 2023; Rothfels, 2023; Wappett et al., 2021; Overton, 2020; Shannon et al., 2003)

It may be beneficial to investigate additional pattern classes within each type of expression table, study other datasets with similar kind of data and analyze relationships between more than two genes.

5.4 FINAL WORDS

To conclude, the findings of the present study demonstrate that the gene dependencies in multiple myeloma samples appear to have a different nature from the one that was expected to be found in the beginning of the actual research. Further analysis and additional computational tools are required to investigate gene dependencies and identify molecular vulnerabilities in multiple myeloma, which could be targeted for the treatment of this cancer.

VI. FUTURE WORK AND LIMITATIONS

6.1 FUTURE RESEARCH

The next steps could include the annotation and enrichment of gene lists obtained from the dataset using DAVID, KEGG, and Reactome databases. NetNC, Cytoscape, and SynLeGG are well-suited for further exploration of network regulation and uncovering gene dependency relationships within gene and gene pair lists. (Sherman et al., 2022; Kanehisa, 2023; Rothfels, 2023; Wappett et al., 2021; Overton, 2020; Shannon et al., 2003)

Two main routes can be highlighted in the advanced analysis of the obtained information. First, complete the finalization of functional annotation and enrichment analysis of single genes and gene pairs to identify the pathways and biological processes associated with their expression. Initially, this study can be performed in DAVID (Database for Annotation, Visualization, and Integrated Discovery) for single genes; additionally, KEGG (Kyoto Encyclopedia of Genes and Genomes) is suitable for understanding high-level functions and utilities of biological systems through pathway mapping of both individual genes and gene pairs from the resulting lists. The outcomes from DAVID and KEGG may then be confirmed using the Reactome database, which is a curated resource dedicated to pathways and reactions in human biology. (Sherman, 2022; Kanehisa, 2023; Rothfels, 2023)

Tools that assist in analyzing network regulatory interactions and gene dependencies include NetNC, SynLeGG, and Cytoscape. All three tools provide visualization of biological network interactions. NetNC may clarify the mutual inhibition regulatory patterns observed during the current study regarding interactions between oncogenes and tumor suppressors. NetNC is suitable for validating the obtained patterns, as it can create modules based on the correlations between the expression levels of genes across samples, suggesting that they may be involved in similar biological processes and represent gene regulatory networks or pathways. The inquiry regarding synthetic lethality dependency can be addressed using both the NetNC and SynLeGG tools. Cytoscape can be utilized for the analysis of both single genes and gene pairs, providing interactive exploration and detailed graphical representations. (Wappett et al., 2021; Overton, 2020; Shannon, 2003)

6.2 LIMITATIONS

The study primarily focused on transcription level; however, additional data could provide valuable insights due to the biological complexity of regulation within cancer pathology. Factors such as sample quality, sample heterogeneity, and potential biases during data collection could have an impact on the result of the study.

It is important to note that the present investigation was limited to synthetic lethality relationships within gene pairs. Existing literature suggests that synthetic lethality can arise from interactions involving more than two genes (Wappett et al., 2021; Wappett et al., 2016). Therefore, restructuring the current dataset or developing a new one that incorporates additional data could

yield more comprehensive information regarding possible synthetic lethality relationships in multiple myeloma samples.

Moreover, it is acknowledged that not all patterns' classes inside each table type were explored, highlighting the necessity for further analysis of the datasets or restructuring of the studied dataset to enhance the outcome from the data investigation.

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<https://themmrf.org/>

<https://www.overton-lab.uk/app/synlegg/#tuid>

APPENDICES

CODES.

Preliminary analysis (with output).

```
> categories <- unique(pattern$Cluster_Combination) #number of table cell
positions types in different types of tables

> numberOfCategories <- length(categories)

> categories
[1] "1_x_2" "1_x_3" "2_x_1" "3_x_1" "1_x_4" "3_x_2" "2_x_2" "1_x_5" "2_x_3"
[10] "3_x_3" "1_x_1" "3_x_4" "4_x_3" "4_x_1" "2_x_4" "4_x_4" "4_x_2" "5_x_1"
[19] "5_x_2" NA      "2_x_5" "3_x_5" "5_x_3" "5_x_4" "4_x_5" "5_x_5"

> numberOfCategories
[1] 26

> categories_1 <- unique(pattern$TotalCluster_Combinations) #number of table
types

> numberOfCategories_1 <- length(categories_1)

> categories_1
[1] 9 15 12 6 20 25 10 4 8 16 NA

> numberOfCategories_1
[1] 11

> print(sort(categories_1))
[1] 4 6 8 9 10 12 15 16 20 25 #more simple visualisation of the table
types

> print(sort(categories)) #more simple visualisation of the cell table type
list
[1] "1_x_1" "1_x_2" "1_x_3" "1_x_4" "1_x_5" "2_x_1" "2_x_2" "2_x_3" "2_x_4"
[10] "2_x_5" "3_x_1" "3_x_2" "3_x_3" "3_x_4" "3_x_5" "4_x_1" "4_x_2" "4_x_3"
[19] "4_x_4" "4_x_5" "5_x_1" "5_x_2" "5_x_3" "5_x_4" "5_x_5"

> scf_count <- sw_pattern %>% group_by(Gene1_ClusterCount, Gene2_ClusterCount,
TotalCluster_Combinations) %>% summarise(count=n()) %>%
arrange(TotalCluster_Combinations)

> scf_count %>% print(width=Inf)
# A tibble: 10 × 4
# Groups:   Gene1_ClusterCount, Gene2_ClusterCount [10]
  Gene1_ClusterCount Gene2_ClusterCount TotalCluster_Combinations count
  <int> <int> <int> <int>
1 1 2 4 87471
2 2 2 6 2231363
3 2 4 8 2246707
4 3 3 9 40792519
5 2 5 10 620321
6 3 4 12 84049458
```

7	3	5	15 18538763
8	4	4	16 45105650
9	4	5	20 20387801
10	5	5	25 2603846

Cluster Combination count.

Example of code for cluster combination count “by hand”.

```
> pattern_2_2_4_1x2 <- pattern %>% filter(Gene1_ClusterCount == 2,
Gene2_ClusterCount == 2, TotalCluster_Combinations == 4, Cluster_Combination
== "1_x_2", New_q_value <= 0.05)> print(nrow(pattern_2_2_4_1x2))
```

Code for automatization.

```
> cl_comb <- pattern_new %>% group_by(Gene1_ClusterCount, Gene2_ClusterCount,
TotalCluster_Combinations, Cluster_Combination) %>% summarise(number = n())
```

Table type standardization (with output).

```
> pattern <- read.table(
(file="../gdr/Combined_Binomial63million_padjusted_qvalLt0.05.txt", head =
TRUE)
> pattern_new <- na.omit(pattern)
> library(dplyr)
> library(tidyr)
> library(stringr)
> pattern_new <- pattern_new %>% separate_wider_delim(Cluster_Combination,
delim = "_x_", names = c("first_cluster_id", "second_cluster_id"), cols_remove
= FALSE)
> pattern_new <- pattern_new %>% mutate(gene1_new = if_else(condition == TRUE,
Gene2, Gene1), gene1_cc_new = if_else(condition == TRUE, Gene2_ClusterCount,
Gene1_ClusterCount), Gene2 = if_else(condition == TRUE, Gene1, Gene2),
Gene2_ClusterCount = if_else(condition == TRUE, Gene1_ClusterCount,
Gene2_ClusterCount), ClusterComb_new = if_else(condition == TRUE,
second_cluster_id, first_cluster_id), second_cluster_id = if_else(condition ==
TRUE, first_cluster_id, second_cluster_id))
> pattern_new <- pattern_new %>% select(-Gene1, Gene1 = gene1_new, -
Gene1_ClusterCount, Gene1_ClusterCount = gene1_cc_new, -Cluster_Combination, -
condition, -first_cluster_id, first_cluster_id = ClusterComb_new)
> pattern_new <- pattern_new %>% unite("Cluster_Combination",
c(first_cluster_id, second_cluster_id), sep = "_x_")
> pattern_new <- pattern_new %>% relocate(Gene1)
> pattern_new <- pattern_new %>% relocate(Gene1_ClusterCount, .after = Gene2)
> pattern_new <- pattern_new %>% relocate(Cluster_Combination, .after =
TotalCluster_Combinations)
> Switched_Pattern <- write.csv(pattern_new, row.names=TRUE,
file="/home/svetlana/switch/Switched_Pattern.txt")
> pattern_new %>% print(width = Inf)
# A tibble: 216,663,899 × 12
  Gene1      Gene2      Gene1_ClusterCount Gene2_ClusterCount
  <chr>      <chr>              <int>              <int>
1 ENSG00000081177 ENSG00000132676          3              3
2 ENSG00000081177 ENSG00000132676          3              3
3 ENSG00000081177 ENSG00000132676          3              3
```

```

4 ENSG00000081177 ENSG00000132676 3 3
5 ENSG00000081177 ENSG00000132680 3 5
6 ENSG00000081177 ENSG00000132680 3 5
7 ENSG00000081177 ENSG00000132680 3 5
8 ENSG00000081177 ENSG00000132680 3 5
9 ENSG00000081177 ENSG00000132740 3 4
10 ENSG00000081177 ENSG00000132740 3 4
  Actual_Count Expected_Count TotalClusterCombinations Cluster_Combination
      <int>      <dbl>      <int> <chr>
1      126      166.      9 1_x_2
2       14      39.3      9 1_x_3
3      171      226.      9 2_x_1
4        2      13.1      9 3_x_1
5       55      89.7     15 1_x_3
6       25      45.7     15 1_x_4
7       71      144.     15 2_x_1
8        0       8.35     15 3_x_1
9       87      137.     12 1_x_2
10      21      45.1     12 1_x_3
  Patients_Count p_value q_value New_q_value
      <int>      <dbl>      <dbl>      <dbl>
1      928 2.06e- 4 1.20e- 2 1.87e- 2
2      928 2.22e- 6 1.90e- 4 3.12e- 4
3      928 9.82e- 6 7.36e- 4 1.19e- 3
4      928 1.98e- 4 1.15e- 2 1.80e- 2
5      928 2.64e- 5 1.81e- 3 2.89e- 3
6      928 4.66e- 4 2.54e- 2 3.94e- 2
7      928 3.37e-13 1.79e-10 3.11e-10
8      928 2.28e- 4 1.31e- 2 2.05e- 2
9      928 7.65e- 7 7.26e- 5 1.21e- 4
10     928 3.51e- 5 2.36e- 3 3.75e- 3

```

Creating combinations table (with example of output).

```

> Comb <- sw_pat %>% select(Gene1, Gene2, Gene1_ClusterCount,
Gene2_ClusterCount, TotalClusterCombinations, Cluster_Combination) %>%
distinct() %>% group_by(Gene1, Gene2, Gene1_ClusterCount, Gene2_ClusterCount,
TotalClusterCombinations) %>% arrange(Cluster_Combination) %>%
summarise(Result_Comb = str_c(Cluster_Combination, collapse = ".")) %>%
arrange(nchar(Result_Comb), Result_Comb) %>% ungroup()

> Comb %>% print(width=Inf, 1:5,)
# A tibble: 53,996,489 × 6
  Gene1      Gene2      Gene1_ClusterCount Gene2_ClusterCount
  <chr>      <chr>      <int>      <int>
1 ENSG00000002586 ENSG00000136021 5 5
2 ENSG00000002586 ENSG00000249855 5 5
3 ENSG00000006327 ENSG00000135828 5 5
4 ENSG00000006327 ENSG00000153140 5 5
5 ENSG00000006327 ENSG00000185220 5 5
6 ENSG00000006327 ENSG00000185875 5 5
7 ENSG00000006327 ENSG00000186106 5 5
8 ENSG00000006327 ENSG00000274349 5 5
9 ENSG00000006652 ENSG00000130772 5 5
10 ENSG00000006652 ENSG00000132763 5 5
  TotalClusterCombinations Result_Comb

```


	<int>	<chr>
1	25	1_x_1
2	25	1_x_1
3	25	1_x_1
4	25	1_x_1
5	25	1_x_1
6	25	1_x_1
7	25	1_x_1
8	25	1_x_1
9	25	1_x_1
10	25	1_x_1

Creating combination tables for each table type (example for 4-cell table).

```
> comb4 <- Comb %>% filter(TotalCluster_Combinations == 4) %>%
  arrange(nchar(Result_Comb))
```

Combination count for each table type (example for 4-cell table).

```
> comb4_n <- comb4 %>% group_by(Result_Comb) %>% summarise(count=n()) %>%
  arrange(desc(count))
> length(comb4_n)
[1] 2
> comb4_n
# A tibble: 6 × 2
  Result_Comb count
  <chr>      <int>
1 1_x_2.2_x_1 35925
2 2_x_1       604
3 1_x_2       601
4 1_x_2.2_x_2 33
5 2_x_1.2_x_2 14
6 1_x_1.2_x_2 1
```

Creating gene pairs and single gene list (with output).

```
> genes_pairs_4 <- comb4 %>% filter(Result_Comb == "2_x_1")
> genes_pairs_4[1:3,]
  x      Gene1      Gene2 Gene1_ClusterCount Gene2_ClusterCount
1 602 ENSG00000008838 ENSG00000021776          2              2
2 603 ENSG00000008838 ENSG00000048991          2              2
3 604 ENSG00000008838 ENSG00000068394          2              2
  TotalCluster_Combinations Result_Comb
1              4          2_x_1
2              4          2_x_1
3              4          2_x_1
> length(genes_pairs_4$Result_Comb)
[1] 604
> unique_gp_4 <- genes_pairs_4 %>% distinct(Gene1, Gene2)
> unique_gp_4[1:3,]
  Gene1      Gene2
1 ENSG00000008838 ENSG00000021776
2 ENSG00000008838 ENSG00000048991
3 ENSG00000008838 ENSG00000068394
```

```

> length(unique_gp_4$Gene1)
[1] 604
> length(unique_gp_4$Gene2)
[1] 604
> gene_list_4 <- unique_gp_4 %>% pivot_longer(c(Gene1, Gene2), values_to =
"Genes") %>% select(-name) %>% distinct(Genes) %>% arrange(Genes)
> head(gene_list_4)
# A tibble: 6 × 1
  Genes
  <chr>
1 ENSG00000008838
2 ENSG00000021776
3 ENSG00000048991
4 ENSG00000056291
5 ENSG00000061337
6 ENSG00000066336
> length(gene_list_4$Genes)
[1] 486
> gene_list_4 %>% print(width=Inf, 1:3)
# A tibble: 486 × 1
  Genes
  <chr>
1 ENSG00000008838
2 ENSG00000021776
3 ENSG00000048991
4 ENSG00000056291
5 ENSG00000061337
6 ENSG00000066336
7 ENSG00000068394
8 ENSG00000074211
9 ENSG00000077522
10 ENSG00000079819

```

PATTERNS VISUALISATION. Visualization of the most important classes of patterns for every table type.

Table 4 (2x2)

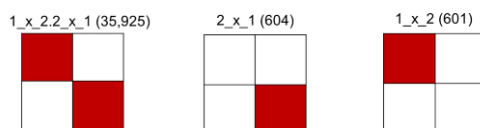


Table 6 (2x3)

1_x_3_2_x_1 (561,010)

1_x_2_1_x_3_2_x_1 (170,230)

2_x_1 (78,768)

Table 8 (2x4)

1_x_3_2_x_1 (370,174)

1_x_3_1_x_4_2_x_1 (201,125)

2_x_1 (166,743)

Table 9 (3x3)

1_x_2_1_x_3_2_x_1_3_x_1 (5,935,212)

1_x_3_2_x_1_3_x_1 (1,058,576)

1_x_2_1_x_3_3_x_1 (953,487)

Table 10 (2x5)

1_x_4_2_x_1 (68,478)

2_x_1 (54,806)

1_x_3_1_x_4_2_x_1 (31,713)

Table 12 (3x4)

1_x_3.1_x_4.2_x_1.3_x_1 (5,397,851)

1_x_3.2_x_1.3_x_1 (4,517,289)

1_x_2.1_x_3.2_x_1.3_x_1 (3,508,021)

Table 15 (3x5)

1_x_3.1_x_4.2_x_1.3_x_1 (1,353,546)

1_x_3.2_x_1.3_x_1 (436,924)

1_x_4.2_x_1.3_x_1 (307,104)

Table 16 (4x4)

1_x_3.3_x_1 (1,306,128)

1_x_3.1_x_4.2_x_1.3_x_1 (1,196,512)

1_x_2.1_x_3.3_x_1.4_x_1 (961,519)

Table 20 (4x5)

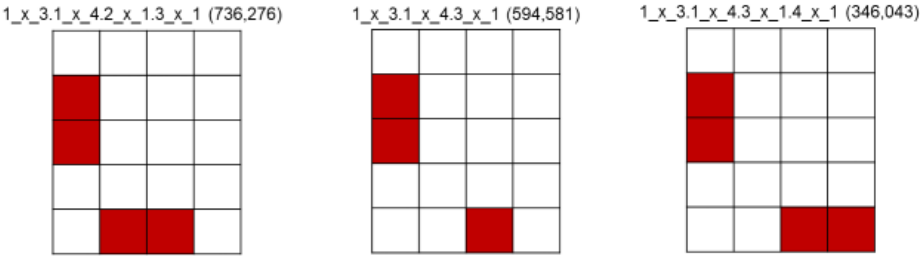


Table 25 (5x5)

